

# The surreal and surcomplex reports

Sixty-three mathematical research packages,  
on the surreal field, its algebraic closure, its quaternion algebra,  
their reach into physics, and the foundations and computation of all of them

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## Scope

This document catalogues the sixty-three research packages stored under `docs`. Each entry names the problem the report attacks, the result it claims, and the delivery it arrived in.

The collection divides in five, and so does the directory tree. Twenty-three reports, under `docs/surreal`, are about the surreal field **No** itself and its real Hahn subfields: birthdays and their behaviour under multiplication, the option graphs of surreal forms, genetic functions and gaps, two questions of Lipparini about evaluating formal sums, the rigidity of exponential automorphisms, Gamma functions, differential transcendence of explicit surreals, Hahn-valued measures and probability, Markov generators, transcendence over bounded-support series, diagonal matrix scaling, vector and tensor fields, Euclidean three-space and its rotations, and finite probability, independent surreal copies, and six reports on Conway’s omnific integers **Oz**: their Diophantine geometry, their set-sized quotients, the automorphisms that preserve them, their algebraic groups and lattices, and the discrete initial subgroups they normalize, and continued fractions with omnific digits. Twenty-eight, under `docs/surcomplex`, are about the algebraic closure **No**[ $i$ ]: the theory of holomorphic functions over it, its differential algebra, its rank-one analytic geometry, its finite and infinite-dimensional spectral theory, its local, quasi-periodic and expanding dynamics, its nonabelian support obstructions, its entire functions at arbitrary valuation rank and their holonomic rigidity, Tate curves and multiscale theta series, Herglotz positivity, moment reconstruction, the summability of Wick expansions, the duals of Hahn vector spaces, the automorphisms of the field, cardinally bounded Hahn fields, what one dilation defines, the Gamma and zeta functions, hidden negative directions of Hermitian forms, Hahn–Hilbert geometry, and algebraic relations between a series and its exponent dilate. One, under `docs/surquaternions`, leaves commutativity behind for Hamilton’s division algebra  $\mathbb{H}_{\mathbf{No}}$  over **No**. Two, under `docs/physics`, address physical models: the first asks what surreal scalars do and do not buy in general relativity; the second asks the same of finite quantum operations and gauge models. Neither claims an experimental prediction. The last nine, under `docs/foundations-and-computation`, stand outside all of these: they are about the subject rather than inside it, asking in what foundation this mathematics legally lives, how one computes with it exactly, which surreal numbers are computable at all, how the surreal field of one set-theoretic universe looks from a larger one, which universe of hereditary sets a birthday cutoff remembers, which surreals and omnific integers are definable, how omnific integers can be named exactly, and how large-cardinal embeddings act on normal forms, and which restricted exponential languages over omnific integers are decidable.

These reports were previously part of a larger collection, in a repository whose subject was elsewhere; they were moved here with their git history. That is why an entry may refer to a merge, a duplicate sweep or a delivery number: those describe how the report reached its present form, and the record is kept rather than rewritten.

Each directory holds a  $\text{\LaTeX}$  source, a `README.md` and a typeset article. Most include verification code and its recorded output. Source manuscripts for merged reports remain in Git history; their provenance notes record contributions and editorial choices. These are AI-assisted research drafts. Some claims and prerequisites have checked Lean proofs, mapped precisely in `FORMALIZATION.md`; no complete verification of the collection is claimed. This catalogue describes the reports rather than certifying them. The cross-report `NOTATION.md` explains shared names

and distinctions between local conventions.

Rebuild this document with

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latexmk -pdf manifest.tex && latexmk -c manifest.tex
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## 1 The surreal field

Twenty-three reports about **No** and its real Hahn subfields. Two questions of Lipparini on evaluating formal sums; two independent halves of Gonshor’s product-birthday conjecture, on domains neither of which contains the other; the subgraph question from Roughan’s paper on birthday arithmetic; a genetic counterexample separating pointwise derivatives from differences of constants; a rigidity theorem for exponential automorphisms; a sharp convexity classification of surreal Gamma functions; differential transcendence of explicit surreals; Hahn-valued measures and probability; Markov generators at every valuation scale; transcendence over bounded-support series; diagonal matrix scaling; vector and tensor fields in every dimension; Euclidean three-space and  $\mathrm{SO}(3, \mathbf{No})$ ; and finite surreal probability; and six reports on the omnific integers **Oz**, written from batches 24–29: Diophantine geometry, set-sized quotients, preserving automorphisms, groups and lattices, and discrete initial subgroups with their omnific normalization, together with continued fractions. A further report proposes independent copies with prescribed Hahn cores and arithmetic-intersection gaps. Among the first sixteen, the first and the fourth each arrived as a pair of archives and were merged; the exponential-automorphism report later absorbed two independent re-derivations of its own theorem, and the measures report merges five manuscripts, the vector-field report three and the three-space report four; individual entries record further expansions. The five earlier omnific reports are merges, grown by later additions: the Diophantine report assembles fifteen manuscripts, the quotient report twenty-four, the preserving-automorphism report fifteen, the groups and lattices report five and the initial-subgroup report two. The continued-fraction report is a written single-source report. Assembly, independent proof review and Lean coverage are separate; the ledger maps the ring and constant-term package, degree laws, units, finite elements and exact omnific floor and ordinary polynomial-root rigidity of the Diophantine report to Lean.

### 1. A square-root obstruction to evaluating increasing Hahn series at $\omega$

[surreal/hahn-evaluation-at-omega/article.pdf](#)

from `hahn_evaluation_research.zip` + `reversed_hahn_series.zip`

Answers Problem 7.7 (p. 38) of Lipparini (arXiv:2505.00424v2) negatively: there is no ring homomorphism from the field of set-supported real-coefficient Hahn series with strictly increasing surreal exponents into the surreals sending  $x^a$  to  $\omega^a$ . The certificate is  $B \in \mathbb{Q}[[x]]$  with  $B^2 = 1 - x$ , built by the recursion  $b_n = -\frac{1}{2} \sum_{k=1}^{n-1} b_k b_{n-k}$ : any such map would force a square to equal  $1 - \omega < 0$ . The homomorphism is applied only to the finite identity, so no continuity, valuation, order preservation or preservation of infinite sums is assumed. Adds the exact classification of admissible images of the variable as the infinitesimals, the monomial classification over an arbitrary ordered abelian group, and order-uniqueness on the source field. *Merged; the core certificate was identical in both originals and is proved once, but the two toolkits are kept — in particular a direct nonnegative-coefficient semiring certificate. Restriction and extension by differences relate the ring and semiring maps; the two witnesses give distinct elementary proofs.*

### 2. Gonshor’s Product-Birthday Conjecture for ordinal-support series

[surreal/gonshor-product-birthdays/surreal\\_product\\_birthdays.pdf](#)

from `gonshor_product_birthdays_research.zip`

Proves Gonshor’s bound  $b(xy) \leq b(x) \otimes b(y)$  on the ring of Conway normal forms  $\sum_{\alpha \in S} c_\alpha \omega^{-\alpha}$  with  $S$  an arbitrary set of ordinals, so transfinite and uncountable supports are allowed. Supplies an exact birthday formula by transfinite induction over the support with maintained prefix bounds and an explicit limit stage, and a complete classification of the equality cases. The route is a leading-Cantor-degree gap, which also gives that the degree of the birthday never rises

under multiplication. The conjecture for arbitrary surreal numbers is not solved. The domain is incomparable with that of the Laurent-series report `surreal/gonshor-laurent-birthdays`: it excludes  $\omega$  and every positive power.

### 3. A Laurent-series case of Gonshor’s birthday conjecture

`surreal/gonshor-laurent-birthdays/article.pdf`

from `Gonshor_Laurent_Birthday_Research.zip`

Proves the same bound  $b(xy) \leq b(x) \otimes b(y)$  on the canonically embedded field  $\mathbb{R}((\omega^{-1}))$ , which unlike the ring of `surreal/gonshor-product-birthdays` contains  $\omega$  and its positive powers but restricts supports to order type at most  $\omega$  on the integer lattice. The route is a bilinear budget: an exact splitting  $b(x) = b(P) \oplus b(u)$  into polynomial and infinitesimal parts, three separate product estimates for the resulting cross terms, and a finite-jet stabilisation carrying the finite bound to infinite series without any continuity claim. Also treats finite normal forms with arbitrary ordinal exponents, derives exact birthdays for finite and infinite Laurent series, and proves a reciprocal corollary. Neither this theorem nor that one contains the other; on their common corner  $\mathbb{R}[[\omega^{-1}]]$  the two agree and are proved twice.

### 4. Canonical forms need not be subgraphs

`surreal/canonical-forms-need-not-be-subgraphs/surreal_graphs.pdf`

from `surreal_graphs_research.zip` + `surreal_sparse_graphs.zip`

Answers negatively the question in the Figure 3 caption of Roughan, *Surreal Birthdays and Their Arithmetic* (arXiv:1810.10373v2, p. 7): the canonical form of a surreal value need not be a subgraph of every other form of that value. Five vertices and rank three are both least possible, and there are exactly sixteen five-vertex counterexamples, eight triangle-free, four of value  $1/2$ . Adds cycle representations of  $1/2$  at every admissible length, the exact extremal function, a planar construction of maximum degree three and arbitrarily large girth, classifications of which canonical graphs are unavoidable, and transfinite witnesses. *Merged; the two originals exhibited genuinely different five-vertex witnesses — one passing through a vertex of value  $-1$ , one with no negative value at all — and both are printed in full, as are two odd-cycle families, two high-girth constructions and two proofs of the Fibonacci bound.*

### 5. A countable genetic cut across the finite–infinite gap

`surreal/genetic-gaps-and-primitives/article.pdf`

from `genetic_gap_counterexamples.zip`

With  $q(t) = t/(1+t^2)$  and  $B(x) = 1 - \{q(n-x) : n \in \mathbb{N} \mid \}$  as a Conway cut, proves that  $B$  is the indicator of the positive infinite surreals, is genetic under the set-sized option constructors of the cited sources, and has pointwise derivative zero everywhere. Consequently  $x$  and  $x - B(x)$  are genetic primitives of 1 agreeing at every real input that do not differ by a constant, the latter also an increasing bijection. The sublevel class  $\{x : 0 < x < \omega, B(x) \leq \frac{1}{2}\}$  is nonempty and bounded yet has no surreal supremum, so neither end gap supplies one.

### 6. A Number-Valued Broadcast Sum of Surreal Sequences

`surreal/broadcast-sum-of-surreal-sequences/article.pdf`

from `surreal_broadcast_sum.zip`

Analyses the sign-truncation game of Lipparini (arXiv:2505.00424v2, Remark 7.6(3), in connection with Problem 7.5). Proves the game always has a surreal-number value, that it agrees with finite surreal sums and with Lipparini’s ordinal operation on their respective domains, and that it fails weak coordinatewise monotonicity, with an explicit family  $s[n] = 3 \cdot 2^{-n-d-1}$  on a subset  $E$  and  $2^{-n-d}$  off it realising the failure. The underlying rule is Lipparini’s; the name and the notation are local to the report.

## 7. Cofinal displacement and the rigidity of exponential automorphisms

[surreal/exponential-automorphism-rigidity/article.pdf](#)

from three manuscripts, delivered 21 and 22 September 2026

Let  $F$  be an ordered field with a total increasing exponential  $E : (F, +) \rightarrow (F_{>0}, \cdot)$  that is *onto*, and  $w$  a nontrivial convex valuation — no Hahn workspace, completeness, saturation or set-size hypothesis. The engine is one finite lemma: the displacement  $D(x) = \sigma(x) - x$  of a unital field endomorphism is *not* a derivation but obeys  $D(ab) = \sigma(a)D(b) + bD(a)$ , and from  $D(a) = d \neq 0$  the two probes  $b = 2B(1 + |\sigma(a)|)/|d|$  and  $ab$  force  $\max\{|D(b)|, |D(ab)|\} > B$ , with neither surjectivity nor order preservation of  $\sigma$ . Through the surjective, additive, order-reversing map  $\nu = w \circ E$ , a nonidentity exponential automorphism stabilizing  $\mathcal{O}_w$  displaces value-group elements cofinally and coinicially, so  $\rho_w : \text{Aut}_{\text{exp}}(F; \mathcal{O}_w) \rightarrow \text{Aut}(\Gamma, +, <)$  is injective; coarsening rigidity takes stabilization of  $\Delta$  as an explicit hypothesis. The kernel of  $\nu$  is a proper convex subgroup, bounded by a *field* element (the finite surreals by  $\omega$ ), so on **No**, with  $B = \omega$ , an exponential automorphism fixing every natural-valuation value is the identity (Theorem 5.1; the hypothesis is not to be weakened to preserving comparisons), hence so is every exponential 1-automorphism, with no strong-additivity or coefficient-fixing hypothesis, offered as a negative answer to Question 5.4 of Kaplan–Krapp–Serra, arXiv:2509.22374v3. *Keep that version pinning prominent*: the reading was checked against that PDF on 22 September 2026, which verifies the interpretation, not priority, and Gonshor’s exponential and the natural valuation are cited, not reproved. Section 8 gives two disjoint families of value-group automorphisms, bounded-layer reweightings and positive rational dilations, that lift as ordered field automorphisms but not exponentially, so cofinal displacement is necessary and not sufficient; Section 9, for real closed  $F$ , gives  $\text{Aut}_L(F(i)) \cong \text{Aut}_{\text{exp}}(F) \times C_2$  for  $L(z) = \log \sqrt{z\bar{z}}$ , with no complex-logarithm branch. Two later manuscripts re-derived the engine and the answer by the same proof, printed once; merging them added the theorem that no nonzero twisted derivation of a nontrivially valued field has all output valuations bounded below by one element, at any rank and with no ordering (Theorem 3.5), rigidity from one visible exponential scale without surjectivity of  $E$  (Theorem 4.4), an order-free detector (Corollary 4.5), profile rigidity on a final interval (Theorem 6.1), cofinal multiplicative motion and cofinal commutation defects of value-fixing maps, the latter not asserting that such a map exists (Section 7), no global bound  $v(\partial y) \geq v(y) + \gamma$  for a nonzero  $E$ -compatible derivation of a nontrivially valued field, applied to the Berarducci–Mantova derivation (Theorem 10.3), and  $\mathbb{R}((t))$  sharpness examples that are not counterexamples (Section 11). Rigidity here means faithfulness, not triviality; the image of  $\rho_v$  on **No** is not classified (Question 13.1); the derivation theorems exclude only global bounds. The Berarducci–Mantova derivation is an object, not a theorem, shared with [surreal/gamma-functions](#) and [surreal/tail-spans-and-differential-transcendence](#), and this third “rigidity” is neither the all-scale polynomial rigidity of [surcomplex/analysis](#) nor the scalar rigidity of [surreal/gamma-functions](#). The staged source audit’s conclusion that the collection lacked this report is false, as its appended correction says. Generic displacement, exponential-profile, logarithmic-modulus and valuation-kernel proofs now have Lean mappings; actual surreal exponential instantiations remain pending. Not refereed; no exhaustive priority claim.

## 8. Gamma functions over the surreals: a sharp convexity classification

[surreal/gamma-functions/article.pdf](#)

from one manuscript, delivered 21 September 2026

Write  $M(x)$  for the leading Conway monomial of a positive infinite surreal with its real coefficient stripped and  $\text{pp}(x)$  for the purely infinite part; for an arbitrary class assignment  $a$  on positive infinite monomials, no regularity assumed, set  $h_a(x) = a(M(x)) \text{pp}(x)$  and  $L_a = L_0 + h_a$  over the Taylor–Stirling log-Gamma baseline. The main theorem is a two-sided sharp threshold:  $L_a$  is convex on all of **No**<sub>>0</sub> iff  $ma(m)$  is finite for every positive infinite monomial  $m$ , and in that case it is *strictly* convex — pointwise in  $m$ , no uniform real bound across monomials, both signs allowed, and no merely-convex member exists. Every member of the family, convex or not, preserves  $\Gamma_a(1) = 1$ , the recurrence  $\Gamma_a(x+1) = x\Gamma_a(x)$ , every finite Gauss multiplication formula, every normalized finite-translation germ and all positive-order fine derivatives of log-Gamma, so the

literal surreal Bohr–Møllerup conditions do not give uniqueness:  $a_\lambda(m) = \lambda e^{-m}$ ,  $\lambda \in \mathbb{R}$ , gives pairwise distinct strictly log-convex extensions whose logarithmic and relative differences are smaller than *every* ordinary inverse power at *every* infinite argument, the log-difference at  $\omega$  being exactly  $\lambda \omega e^{-\omega}$ , and adding all ordinary-order Stirling envelopes does not restore uniqueness. Absolute Gamma differences need not be small: at  $\omega$  the difference  $\Gamma_1 - \Gamma_0$  exceeds every ordinary power of  $\omega$ . The convexity proof uses no second derivatives and no mean value theorem: it works with the tangent gap  $D_F(y, x) = F(y) - F(x) - F'(x)(y - x)$  through four exhaustive relative-scale regimes, the margin  $(y - x)^2/(2x)$  being exactly what generates the  $O(m^{-1})$  threshold, and the mechanism is then isolated as a reusable scale-margin gauge principle. A positive second derivative does *not* imply convexity in this calculus — the gauge with value one on one monomial and zero elsewhere has all fine derivatives equal to the baseline’s and a negative tangent gap — because such gauges are finely locally constant and convexity cannot be decided locally. Compatibility with the Berarducci–Mantova derivation forces  $h_a$  into  $\ker \partial = \mathbb{R}$  and kills every member but  $a = 0$ , yet a different real-valued non-flat gauge survives, so *scalar compatibility alone is not a uniqueness characterization*. Closes with an exact finite-phase domain on the tube  $\operatorname{Re}(z) > 0$ ,  $\operatorname{Im}(z)$  finite, using the finite-angle exponential of the imaginary log-Gamma value, distinct from choosing a primitive in **surcomplex/differential-equations**. *Not offered as a resolution of a named published problem*, and the classification is exact only for the named monomialwise family. Its own audit read an earlier state of this catalogue; its eighteen-package count is a historical description, not the current inventory.

## 9. Tail-spans and differential transcendence in surreal and surcomplex Hahn fields

[surreal/tail-spans-and-differential-transcendence/article.pdf](#)

from one manuscript, delivered 22 September 2026

Let  $M$  have characteristic zero,  $\Gamma$  be a set-sized ordered abelian group,  $(a_i) \subset M^\times$  have independent square classes with  $r_i^2 = a_i$ , the exponents  $\gamma_i$  be distinct with well-ordered image, and  $v_i \in M^r$ ; put  $Y = \sum_i t^{\gamma_i} r_i v_i$  over  $B = M((t^\Gamma))$ . The invariant is the *cofinite span*  $W$ , the intersection of the spans surviving every *finite* deletion, not deletion of a transfinite initial segment. Theorem 5.1:  $E = \{i : v_i \notin W\}$  is finite and  $B[Y_1, \dots, Y_r] = B_E[s_1, \dots, s_d]$ , where  $d = \dim_M W$ ,  $B_E = M(r_i : i \in E)((t^\Gamma))$  has degree  $2^{|E|}$  over  $B$  and the  $s_j$  are algebraically independent over  $B_E$  — an equality of subrings, not of dimensions; Theorem 5.2 splits the relation ideal of  $Y$  over  $B$ , after base change to  $B_E$ , into  $2^{|E|}$  pairwise comaximal affine planes of dimension  $d$ . The proof is finitary: the alternating sum of finitely many coefficient sign changes of a degree- $D$  relation is a mixed difference equal to  $D!$  times its top polarization, which cofinite spanning forces to vanish identically. With  $t = \omega^{-1}$ ,  $B_0 = \mathbb{Q}((t^\mathbb{Q})) \subset H_0 = \mathbb{A}_\mathbb{R}((t^\mathbb{Q})) \subset \mathbf{No}$  (real-algebraic coefficients) and  $\xi_A = \sum_{n \in A} \sqrt{p_n} \omega^{-n}$ ,  $p_n$  the  $n$ th prime: for *every* infinite  $A$  the Berarducci–Mantova jets  $\partial^k \xi_A$  are algebraically independent over  $B_0$  (Theorem 7.1), and those of the explicit binary-prefix almost disjoint family  $(\xi_\alpha)_{\alpha \in \{0,1\}^\mathbb{N}}$  jointly (Theorem 7.3); with a separate count  $|H_0| \leq 2^{\aleph_0}$  for the upper bound,  $\operatorname{dtrdeg}_{B_0} H_0 = \operatorname{trdeg}_{B_0} H_0 = 2^{\aleph_0}$ . For every infinite  $A$ , all mixed jets  $\partial_z^j \delta^k F_A$ ,  $\delta = t d/dt$ , of  $F_A = \sum_{n \in A} t^n \sqrt{1 - z/2^n}$  are independent over  $\mathbb{C}(z)((t^\mathbb{Q}))$  (Theorem 9.3), and continuum many such functions jointly on the one disk  $|z| < 1$  (Theorem 10.1). Also a value-group-relative valuation-closure criterion (Theorem 12.1) and undecidability of algebraicity and differential algebraicity over  $B_0$  for total computable coefficient streams (Proposition 13.1). Cautions: the sign changes are not order-preserving and are not extended to  $\mathbf{No}$ ;  $\mathbb{C}(z)((t^\mathbb{Q}))$  is not a common-domain ring; point values are not claimed transcendental;  $F = F_{\mathbb{N}_{>0}}$  satisfies  $F(2z, t) - tF(z, t) = t\sqrt{1 - z}$ , so functional equations are not excluded; the family indexed by *all* subsets is dependent,  $\xi_{A \cup B} + \xi_{A \cap B} = \xi_A + \xi_B$  (Remark 7.4, added on fitting). The formula for  $\partial$  on  $\mathbb{C}((t^\mathbb{Q}))$  is cited from **surcomplex/differential-equations**; only the restriction to real-algebraic coefficients is checked here. **surcomplex/dynamics-and-normal-forms** keeps, unweakened, its warning that its  $H_\varepsilon$ , with  $H'_\varepsilon = \Omega_\varepsilon H_\varepsilon$ , is not differentially transcendental; Theorem 9.3 concerns different functions. Its 404 exact finite checks prove no infinite statement; priority not certified, not refereed. The Lean ledger records proofs of cofinite-span stabilization and multilinear detection; the classification and differential-independence theorems remain pending.

## 10. Hahn-valued measures and probability: strong atomicity, coefficientwise extension, and hidden negative mass

[surreal/hahn-valued-measures-and-probability/article.pdf](#)

from five manuscripts, delivered 22 September 2026

Measures with values in  $\mathbb{R}((t^\Gamma))$  and  $\mathbb{C}((t^\Gamma))$ , and through  $t^\gamma = \omega^{-\gamma}$  in **No** and **No**[ $i$ ], with  $\Gamma$  a set-sized ordered abelian group *not* assumed divisible except in one Gaussian quadrature theorem, under two meanings of countable additivity: a *coefficientwise* Hahn measure is a Hahn series of ordinary finite signed or complex measures on one well-ordered exponent set, added exponent by exponent; a *strong* one requires the masses of every disjoint sequence of events to be strongly Hahn summable. **Strong class.** On every countably separated space a strong measure is  $\mu(A) = \sum_{x \in A}^H w_x$  for one strongly summable family  $(w_x)$ , common support and atomicity being conclusions; it extends uniquely to all subsets, and a strong probability has finitely supported standard part. Exact criteria are proved for coherent cylinder data over finite alphabets, for independent finite-state coordinates (finitely many random at the residue level, and the individual rare entries, not their aggregates, strongly summable), and for moments on an ordinary real sample set. One weight pattern of support order type  $\omega + 1$  gives nonnegative Cantor-space cylinder masses, or a functional strictly positive on nonzero Hahn polynomials nonnegative on  $[0, 1]$ , yet a negative singleton;  $\omega + 1$  is least. **Coefficientwise class.** A transfinite null-ideal criterion decides positivity; two exponents already hide a negative singleton; independent Bernoulli perturbations on a common well-ordered exponent set extend exactly under weighted  $\ell^2$  rows at baselines uniformly separated from 0 and 1, and  $\ell^1$  rows at deterministic ones. *The threshold  $\omega + 1$  and the finitely supported standard part hold in the strong class only.* Sources 10 and 11 proved the classification independently, and the threshold for cylinder and polynomial tests respectively; each is printed once, 10 as base; source 15's classification on compact metric spaces, with a common support *assumed*, is recorded as a special case. Not a first non-Archimedean Kolmogorov theorem (Ludkovsky–Khrennikov proved one under other axioms); priority not certified and not refereed. The Lean ledger records checked strong atomicity and product prerequisites; the complete report is not formalized. [physics/surreal-scalars-and-spacetime](#) asked for a summation theory, and this report works out two; it shows that the object refused by non-claim N75 of [surcomplex/dynamics-and-normal-forms](#) cannot exist in the strong class for an ordinary group action with only infinite orbits, and states canonically the null-ideal criterion proved independently in [surcomplex/hahn-herglotz-positivity](#). A fourth manuscript (Sections 25–31) develops the coefficientwise class for countably many scales: every measure has a Hahn-valued density with respect to one finite real control measure, positivity is positivity of that density almost everywhere (related exactly to the null-ideal criterion, whose proof it re-derives for countable exponent sets), and Hahn–Jordan, Radon–Nikodym, Fubini, conditional expectation and infinite masses follow. A fifth manuscript (Sections 32–34) removes the uniform separation of the Bernoulli product theorem: at arbitrary interior baselines the laws extend exactly when every exponent row lies in  $\ell^1 + \ell^2(1/[p_j(1 - p_j)])$ , answering the report's near-boundary question; a mixed-baseline version is added by the merge and marked as such.

## 11. Surreal Markov generators at every valuation scale

[surreal/markov-generators-at-every-scale/article.pdf](#)

from one manuscript, delivered 22 September 2026

Finite Markov chains with positive Hahn rates  $q_e \in F_{>0}$ ,  $F = \mathbb{R}((t^\Gamma))$ , on a strongly connected digraph with  $n \geq 2$  states;  $\Gamma$  is nonzero, set-sized and *divisible* throughout, read in **No** through  $t^\gamma \mapsto \omega^{-\gamma}$ . The object is the normalized resolvent  $R_L(s) = s(sI + L)^{-1}$  of the row Laplacian  $L$  (generator  $-L$ ), defined rational-algebraically, and its residue matrix  $K_\alpha(c) = \text{res } R_L(ct^\alpha)$  at each scale  $\alpha \in \Gamma$ , the report's *shadow*, unrelated to the shadow topology. In-forests give the valuation and leading coefficient of *every* resolvent entry, zero-residue entries included (Theorem 4.1). There are at most  $n - 1$  crossover scales, and the plateaux form a flag of distinct stochastic idempotents  $I = P_0, \dots, P_m = \mathbf{1}\pi_0$ ,  $P_i P_j = P_j P_i = P_{\max(i,j)}$ , of strictly decreasing rank (Theorem 5.2). On the canonical stochastic retract  $P = UV$ ,  $VU = I$ , each crossover is  $U c(cI + G_\alpha)^{-1} V$  for a

real Markov row Laplacian  $G_\alpha = (VK_\alpha(1)U)^{-1} - I$  with semisimple zero eigenvalue, with no reversibility, simple-eigenvalue or unique-minimizing-forest assumption (Theorem 6.2). Conversely (Theorem 7.4, Corollary 7.5), every such flag ending in rank one, with any compatible real effective generators, occurs at any prescribed scales  $\alpha_1 < \dots < \alpha_m$  in an irreducible Hahn row Laplacian, whose intermediate-scale positivity corrections add no crossover. Also: relative rate errors of valuation  $\geq \delta$  on one fixed graph give the same bound for every resolvent entry, uniformly in  $s > 0$  (Theorem 8.1; sharp by Example 8.3); a real one-parameter monomial model reproduces the whole *finite* marked-forest leading diagram, so such data cannot certify higher rank (Theorem 9.2); and  $\det(cI_p + G_\alpha) = cD_\alpha(c)/b_{k_-}$  yields the leading surcomplex eigenvalue amplitudes of  $L$ , including tied scales and nonreal eigenvalues, while the forest profile is convex (Theorems 10.2–10.3). *One manuscript; no merge*: placement collected its own limitations as (N1)–(N11) and added a divisibility ledger and cross-references, changing no theorem number. Not treated: a global surreal exponential, path measures, infinite state spaces, nondivisible  $\Gamma$ . Classical forest, stochastic-idempotent and rank-one metastability inputs are credited; priority is not certified; not refereed. The Lean ledger records checked algebraic clauses of the resolvent identity, projection-flag construction and irreducible completion; residues and the remaining hierarchy theorems are pending. The finite exact checks are not proofs. Lemma 10.1 is the forest-polynomial case of the Newton rule of `surcomplex/polynomial-algebra`; the positivity mechanism parallels the Cauchy–Binet identity of `surcomplex/spectral-theory` for another matrix class; neither this report nor `surreal/hahn-valued-measures-and-probability` depends on the other.

## 12. Transcendence over bounded-support Hahn series

[surreal/transcendence-over-bounded-support/article.pdf](#)

from one manuscript, delivered 22 September 2026

Let  $G$  be a nonzero set-sized ordered abelian group,  $K$  a field of characteristic zero,  $\mathcal{B}_G(K)$  the ring of Hahn series whose support is bounded above,  $\mathcal{F}_G(K)$  its fraction field and  $\kappa = \text{cf}(G)$ , cofinality in exactly the sense of `surcomplex/entire-functions-at-arbitrary-rank`. There are  $2^\kappa$  elements of  $K((t^G))$ , with positive integer coefficients and one common support of order type  $\kappa$ , algebraically independent over  $\mathcal{F}_G(K)$  (Theorem 5.3); no smaller support cardinality carries even one transcendental element (Corollary 5.4), every valuation ball contains such a family, and for real-exponent Hahn fields over  $\mathbb{R}$  or  $\mathbb{C}$  the transcendence degree is the continuum. A descent theorem makes  $K((t^H))$  and  $\mathcal{F}_G(L)$  linearly disjoint over  $\mathcal{F}_H(K)$  for cofinal  $H \subseteq G$  and  $K \subseteq L$  (Theorem 3.2), so relation ideals and finite degrees survive enlargement of coefficients and cofinal exponents. Also: when  $\mathcal{B}_G(K)$  is a field, its units via the highest Archimedean quotient, a GCD-descent theorem that stays *conditional*, and Conway-normal-form witnesses that are actual surreal numbers. Transcendence here comes from the support, not from the coefficients as in `surreal/tail-spans-and-differential-transcendence`; the two bases do not contain each other, and neither report uses the other. One manuscript; priority not certified, not refereed, no Lean verification.

## 13. Diagonal matrix scaling at surreal scales

[surreal/matrix-scaling-at-surreal-scales/article.pdf](#)

from one manuscript, delivered 22 September 2026

Diagonal scaling of a positive matrix to fixed row and column sums over  $F = \mathbb{R}((t^\Gamma))$  and  $K = \mathbb{C}((t^\Gamma))$ , whose entries may sit at arbitrarily different infinitesimal scales; the local theory assumes no rank or divisibility for  $\Gamma$ . The normalization has a unique support-controlled analytic branch on the whole relative infinitesimal polydisc. The invariant is the *spanning-tree deletion gap*: with  $\tau$  the least valuation sum over spanning trees of the bipartite support graph and  $\tau_e^-$  the least over trees avoiding the edge  $e$ ,  $\kappa_e = \tau_e^- - \tau$  is the exact uniform gain at entry  $e$  — an input change of valuation at least  $\delta > 0$  moves that output entry by valuation at least  $\delta + \kappa_e$ , a nonlinear two-point estimate, attained by changes in  $e$  alone, with bridges fixed exactly. Taylor coefficients obey the same bounds; evaluation at arbitrary rank needs a separate finite-generator support argument; along a tridiagonal bistochastic chain the gains add exactly. The positive-sum

mechanism is that of the forest sums of **surreal/markov-generators-at-every-scale** and the Cauchy–Binet sums of **surcomplex/spectral-theory**, but the maps differ: the Markov report’s stability bound has no gain, and  $\kappa_e$  is not a condition number. No convergence or complexity claim for Sinkhorn iteration. One manuscript; not refereed, no Lean verification, priority not certified.

#### 14. Surreal vector and tensor fields in every dimension

[surreal/vector-and-tensor-fields/article.pdf](#)

from three manuscripts, delivered 22 September 2026

Replacing  $\mathbb{R}$  by **No** gives tensor algebra but no calculus. Three manuscripts built the same layered theory, printed once from the  $n$ -dimensional one: finite algebra over a set-sized real-closed surreal subfield; a sheaf of locally Hahn-supported smooth tensor fields on ordinary real spaces, under one support condition that makes sums, products, contraction, differentiation and integration over compact real chains compatible; and genuine surreal-coordinate neighbourhoods reached by support-controlled Taylor prolongation and scale charts. Part I gives Lie and Cartan calculus, connections and curvature, Hodge theory for a real or regular Riemannian Hahn metric, Stokes and Poincaré, Hahn cohomology, standard-part reduction and an infinite-frequency obstruction. Part II, from the three-dimensional manuscript, is dot and cross products, integral theorems and multiscale continuum equations; Part III, from the Minkowski manuscript, is Lorentz transformations at every scale, the Lorentzian Hodge operator, Maxwell and scalar fields and stress–energy, as mathematics on Minkowski space — not an assessment in the sense of **physics/surreal-scalars-and-spacetime**. Results proved in three dimensions are printed in every dimension only where the proof allows, and nothing Lorentzian is attached to a general statement. The complex-coefficient Stokes and cohomology results of **surcomplex/contours-and-stokes** are credited, not reclaimed. Not refereed, no Lean verification, priority not certified.

#### 15. Euclidean three-space and the rotation group over the surreals

[surreal/euclidean-three-space/article.pdf](#)

from four manuscripts, delivered 22–23 September 2026

Coordinate geometry of **No**<sup>3</sup> inside set-sized real-closed Hahn fields, where lengths, areas and radii may be infinite or infinitesimal but angles between directions always have canonical finite representatives, so spatial trigonometry needs no oscillatory meaning for an infinite angle. Part I, the base: incidence, projections, skew-line distance, sphere intersections, quadrics, rigid motions, Rodrigues rotations, screw motions, spherical coordinates, and spherical trigonometry with sine and cosine laws, branch-correct excess, area and holonomy. Part II is the class group  $\mathrm{SO}(3, \mathbf{No})$ : every nonidentity rotation has an axis, Rodrigues, Euler and Cayley coordinates stay exact, the standard part gives a split extension  $K \rtimes \mathrm{SO}(3, \mathbb{R})$  whose infinitesimal kernel is uniquely divisible and perfect (every element a single commutator), finite subgroups are conjugate to real ones, and the rotation exponential has rank-one critical spheres. What both manuscripts proved (isometries, Rodrigues, the Cayley chart, the finite phase) is printed once; the spin double cover is cited from **surquaternions/surquaternions**, and the plane case is the  $\mathrm{SO}(2, \mathbf{No})$  section of **surcomplex/trigonometry**. Not refereed, no Lean verification, priority not certified. A third manuscript (Part III) classifies the normal subgroups of  $\mathrm{SO}(3, F)$  by valuation cuts, computes lower central and derived series, and proves that every class homomorphism from  $\mathrm{SO}(3, \mathbf{No})$  to a set-sized group factors through standard part, also for  $\mathrm{SO}(n, \mathbf{No})$  and  $\mathrm{SU}(n, \mathbf{No}[i])$ . A fourth manuscript (Part V, 23 September) answers the report’s question on a uniform classification for compact groups: for a connected real linear algebraic group  $G$  with compact connected real points, every class homomorphism from  $G(\mathbf{No})$  to a set-sized group kills the infinitesimals of the derived group, a universal set-sized quotient exists exactly when  $G$  is semisimple and is then standard part, for semisimple  $G$  the infinitesimal normal subgroups are products of valuation-cut subgroups over the simple factors, and  $|\mathrm{SO}_n(\mathbf{Oz})| = 2^{n-1}n!$ ; by contrast every set-sized image of  $\mathrm{SL}_n(\mathbf{No})$ ,  $n \geq 2$ , is trivial.

## 16. Finite surreal probability, log-odds and conditioning

[surreal/finite-surreal-probability/article.pdf](#)

from one manuscript, delivered 22 September 2026

Probability with values in set-sized subfields of **No**, with logarithmic odds from the surreal exponential. On finite event algebras, where finite, strong and coefficientwise additivity coincide: conditioning on infinitesimal events and exact Bayesian updates, a leading-scale calculus with coefficient-sensitive tie breaking, quantitative conditioning stability, softmax models, relative entropy, proper scoring rules, Gibbs variational principles and finite-horizon processes; a conditional real shadow keeps what the unconditional standard part destroys. Beyond finite algebras it gives regular finitely additive laws on all subsets for two coin models, a fine-ultrafilter law, a Loeb bridge, and a counterexample to positive coefficientwise integration of Hahn-valued functions. Its coin theorems are the cases  $p = t$  and  $p = \frac{1}{2} + t$  of the i.i.d. corollary of [surreal/hahn-valued-measures-and-probability](#), which it answers no question of. No physical interpretation is claimed, in line with [physics/surreal-scalars-and-spacetime](#). One manuscript; not refereed, no Lean verification, priority not certified.

## 17. Omnific integers and omnific–Diophantine geometry

[surreal/omnific-diophantine-geometry/article.pdf](#)

from fifteen manuscripts, delivered 22–23 September 2026

What do Diophantine equations look like over Conway’s omnific integers  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$ ? The theory is organized around the ring retraction  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  onto the constant coefficient: an integer system has an omnific solution iff it has an integer one, so Hilbert’s tenth problem over  $\mathbf{Oz}$  is the ordinary one; the finite quotients are the  $\mathbb{Z}/n$ , and linear systems are solved completely. Existence is kept apart from the structure of solutions. At a nonzero level the fibers of products of linear forms are described exactly, so Pell equations for every  $D \neq 0$ , binary forms with two distinct projective factors and norm forms of finite étale algebras gain no new omnific or Gaussian omnific points, while every represented level of a nondegenerate indefinite quadratic form in at least three variables carries injective polynomial families, in an exhaustive quadratic-level dichotomy. Support-preserving Euler derivations make  $ax^m + by^n = c$  ( $abc \neq 0, m, n \geq 2$ ) rigid, and more generally, over the one-sided ring, give only constant points on every smooth affine curve other than the affine line, on nonsingular Weierstrass cubics, abelian and semiabelian varieties, proper varieties with enough symmetric differentials, and logarithmic complements whose logarithmic forms separate tangent directions. The new singular-curve companion claims that a geometrically integral affine curve has nonconstant points exactly when its normalization is the affine line; the criterion now has a manuscript proof review. Pell rigidity with the three- or four-square theorem gives single quartic equations defining  $\mathbb{Z}$  in  $\mathbf{Oz}$ ;  $x^2 = 2y^2$  defines  $\Pi$ , one order-free system defines  $\mathbb{Z}$  in  $\mathbf{Oz}$  and  $\mathbb{Z}[i]$  in  $\mathbf{Oz}[i]$ , and the graph of  $\text{ct}$  is Diophantine, so every homomorphism preserves constant terms. The pure ring  $\mathbf{Oz}$  interprets **No** and its automorphisms extend to maps fixing  $\mathbb{R}$ , while a phase twist shows that  $\mathbf{Oz}[i]$  does not determine the real axis. Fractions: a point with a unimodular presentation has lowest terms unique up to sign or no gcd at all, according as its constant-term specialization is rational, and the specialization fibers are dense orbits of the congruence kernel of  $\text{SL}_2(\mathbf{Oz})$ . *Assembled from fifteen manuscripts through `fdedce0`; the report states which source proved each result and prints shared results once.* A manuscript proof review covers elementary arithmetic, fractions and differential-rigidity prerequisites, then curves through Section 18.6, including groups, coefficient algebras and logarithmic applications. It corrects rational specialization, normalization lifting and superelliptic irreducibility, and expands the conductor and arithmetic proofs. The exact scope and source comparisons are in `RECONCILIATION.md`. Sections 20–21.6 now have a workspace and question-status review; remaining pointers, neighbouring-report comparisons and full source reconciliation remain pending. Section 19.4 treats constant discriminants, translation of positive-degree monic polynomials and units in finite étale algebras. Its elementary lemmas, root-velocity certificate and translation proof through the omnific-root corollaries have been reviewed, with explicit degree and squarefreeness conventions. A further pass covers the factor, Galois and resultant consequences and the finite étale unit theorem, including

its constant subalgebra and reduction fiber. Monogenic and arithmetic descent, critical-point and normal-matrix applications, boundary examples and the splitting-algebra second proof are also reviewed. The later question/status review separates full faithfulness from descent existence, and gives subring inheritance and affine-conjugacy criteria. Full parallel-source reconciliation awaits review. Lean covers the ring and constant-term package, degree laws, units and finite elements, the exact omnific floor, ordinary polynomial-root rigidity, ordinary residue rings and divisors of nonzero ordinary integers, mixed gcds, Chinese remainders and all finite quotients. Not claimed: a classification of affine varieties of higher dimension, primitive non-unimodular Fermat triples, or squarefree rigidity in positive characteristic. Not refereed, priority not certified.

## 18. Set-sized quotients of the omnific integers

[surreal/set-sized-quotients-of-omnific-integers/article.pdf](#)

from twenty-four manuscripts, integrated through ef3d112

Can the omnific integers be seen from a set? For a unital subring  $D$  of a set field  $K$  and  $\Pi_K$  the purely infinite normal forms over  $K$ , every additive multiplicative map from  $D + \Pi_K$  to a set-sized ring, commutative or not, unital or not, kills  $\Pi_K$  and factors uniquely through the constant term. So the only unital map from  $\mathbf{Oz}$  is  $x \mapsto \text{ct}(x)1$ , the set-sized quotients are exactly the  $\text{ct}^{-1}(\mathbf{a})$ , and every set-sized module is killed by  $\Pi$ . The theorem fails in every fixed set-sized workspace and for finite supports, which admit the augmentation, while countable supports already suffice. For five set-sized integer parts inside  $\mathbf{Oz}$ , the least ring or module detecting a purely infinite element has exactly the size of the ring, in ZFC. Inverting a purely infinite element forces a copy of  $\mathbf{No}$ ;  $\mathbf{Oz}/(1 + \omega^a)$  has no nonzero set-sized image yet contains an algebraically closed field over  $\mathbf{No}[i]$  and a Cantor algebra. The integral closure  $\mathcal{N}$  is proper and fine-dense;  $\mathcal{N}/\Pi\mathcal{N}$  is a finite-partition Boolean power of the algebraic integers whose idempotents form an atomless proper class, and no set of maps to set-sized rings is jointly faithful on it. Integer-valued polynomials give the universal set-sized image  $\text{Int}(\mathbb{Z}^r)$  of the numerical polynomials; polynomial images have holes at every fresh scale, and rational bijections are affine. Homologically, over the coordinatewise cores  $A_d$  the module  $D$  has flat dimension  $d$  but projective dimension  $d + 1$ , lattice ideals  $M + \Pi$  are not finitely presented,  $\mathbf{Oz}$  is not coherent, and derived tensor data recover the multiplication of an order. The Grothendieck ring of the ordinals under natural operations is not a quotient of  $\mathbf{Oz}$ , a negative answer to the addendum of Elliott's MathOverflow question 188430. The new finite-test chapter gives criteria using a scale outside the coefficient support group; it awaits proof review. *Assembled from twenty-four manuscripts; where sources prove one theorem under different hypotheses, as for the cardinal thresholds, the most general form is printed and the agreement explained, and the source limitations are kept.* Not claimed: the general fixed-group threshold, the homological dimensions of  $\mathbf{Oz}$  itself, the isomorphism type of the Boolean algebra. The elementary arguments through the universal quotient theorem have been reviewed, with corrected characteristic assumptions for integer divisibility and a separate noncommutative module argument. The review now extends through representations, small quotients, completions, localizations, positive existential conservativity, modules, derivations, the finite-side standard-part theorem, support thresholds and the initial fixed-group bounds, followed by both two-armed models with their exact detection and generator counts. The Gaussian fraction field uses complex coefficients. An explicit singular-support example separates field closure from the common-monomial-divisor property. The all-cardinal controlled model is also reviewed: finite coordinate use preserves small gaps, and a characteristic-zero residue field sends the actual omega to  $-1$ . The countable-support one-arm model now has expanded residue-field and comaximality proofs, with its infinite-cardinal hypothesis explicit; its Jacobson radical is zero despite a nonzero finite residual. The source-11 model and organizing ring/module theorem are reviewed too: the exact small-scale-field counts retain each support bound and coefficient field. The source-11 argument permits finite coefficient fields; omnific embeddings are asserted for the real/integer and Gaussian specializations. The set-sized homological package is now reviewed: a flat idempotent ideal, flat dimension one for the quotient, natural Ext comparison, and exact coefficient-module thresholds in degrees one and two. The existing ring section does not give an  $A$ -linear splitting.

The finite-support cores are reviewed through their exact flat dimensions: the face resolution and Boolean Tor formula distinguish coordinate rank even when finite quotients and positive self-Tor agree. The lattice syzygies and ordered-cone comparisons are reviewed as well. Noncoherence now has the required coefficient-field restriction;  $K[\mathbb{Q}_{\geq 0}]$  is a coherent counterexample to the unrestricted claim. Tensor normal forms through exterior and symmetric powers now include explicit universal properties for the class carriers; torsion over the omnific ring is compatible with freeness over its ordinary constants. The Rees presentation and first nonlinear degree are reviewed too: algebraic degree gives two-generated ideals with arbitrarily late first nonlinear equations. Their relation ideal is finitely generated over the symmetric algebra, but when nonzero is not finitely generated as a module over the omnific ring. Scalar homomorphisms, ideal classes, duals and common-scale matrices are reviewed: coefficient lattices can remain nonisomorphic despite identical biduals, and a common-scale matrix has finitely presented image exactly when its ordinary kernel is defined over the fraction field of the constant ring. In that case the image is free. The set-presentation obstruction and exact relation counts are reviewed as well. In the lexicographic model of regular cardinal  $\kappa$ , every finite generating map onto an ideal from a coefficient lattice of rank at least two has a kernel needing exactly  $\kappa$  generators. The countable model has projective dimensions two for the ideal and three for its cyclic quotient; these counts are distinct from the uncountable quotient-cardinality thresholds. The coefficient-Tor proof now gives its maps explicitly: second Tor is the kernel of ordinary coefficient multiplication. The maximal self-Tor rank detects a field after extending scalars, even when the lattice itself is not a ring. The quadratic free-kernel hypothesis is sufficient, not necessary. The rank-two moduli and endomorphism orders are reviewed as well: fractional-linear changes classify continuum many ideals, while the endomorphism ring recovers an ordinary order by its constant quotient. For the class ring, agreement with a free module on set-sized tensor tests is distinguished from flatness. The face-resolution review now reaches the free telescope and its top Ext formula: a contractible kernel proves exactness without regular Koszul sequences, and dualizing gives an explicit product cokernel. The finite-support obstruction proves nonvanishing and exact projective dimensions, including the lattice quotients by dimension shifting. The stronger global-dimension bounds require  $K \neq \text{Frac } D$ : for  $A_1 = K[\mathbb{Q}_{\geq 0}]$ , weak global dimension is one although  $\text{pd}_{A_1} K = 2$ . The flat DGA and pair-Tor proofs now check the multiplication and differential signs, the coefficient-kernel map and the first-Tor window. That window always contains a nonzero monomial, even when second Tor vanishes; its decomposition is only  $D$ -linear. The subsequent review proves the linear-disjointness criterion and checks the self-Tor product's tensor order and sign. Its alternating submodule and symmetric-relation quotient form an integral exact sequence; equal ranks for orders do not specify a canonical splitting. Framed reconstruction now includes an integral kernel basis and trace calculations. In contrast, the derived constant-term shadow is square-zero and forgets order multiplication. The universal module assertion is explicitly restricted to the lower universe in the two-universe reading; the regular module in the larger universe is a counterexample without it. All 30 standard polyhedral proofs, the lattice comparison and the explicit nonflatness example are reviewed. Radical coefficient-full monomial ideals are classified by face antichains; join homology computes their quotient's exact flat dimension over every unital  $D \subseteq K$ . Nonzero radical geometric ideals give nonflat homological quotient maps. Flat radical geometric ideals recover the marked face lattice, and simpliciality is equivalent to every facet quotient having flat dimension one. Cube facets can instead attain  $d - 1$ . Squarefree Betti multiplicities transfer to the root core. Over a principal ideal domain that is not a field, nonzero arithmetic modules have flat dimension one if torsion and  $d$  otherwise. Lattice quotients give weak global dimension at least  $d + 1$  when  $K \neq \text{Frac } D$ . For  $d \geq 2$ , a two-monomial relation obstructs flatness of the total-order extension and fails the same finite criterion in the full omnific rings. The simplicial projective formula requires a proper face; the identity quotient has flat and projective dimension zero. The ten question/status notes are reviewed: with fixed coefficients and rational space, a full-dimensional pointed rational polyhedral cone inclusion is flat exactly for equal cones. Interior monomials detect every strict enlargement, even when old ray-pair tests pass. A large quotient

may still have small images:  $\mathbf{Oz}/(\omega)$  surjects onto  $\mathbb{Z}$ . Internal-field proofs are reviewed through compression and transfer to residue fraction fields. Inverting a nonzero purely infinite element forces a copy of  $\mathbf{No}$  (or  $\mathbf{No}[i]$  in the Gaussian case). The field embedding has a proper image with an exact support description, and denominator choices give equal transfer maps on overlaps. Binomial kernels and the dyadic algebras are now reviewed: finite interpolation gives the inverse Cantor map, and a three-root idempotent lies outside every dyadic stage. In Gaussian Cantor coordinates, conjugation acts by  $f(z) \mapsto \overline{f(-1-z)}$ . Class-residue foundations are reviewed through binomial maximal extensions and coefficient comparison: finite witnesses give maximal ideals under global choice, and the unrestricted existence principle is equivalent to global choice already for class domains of characteristic two. The cutoff, tail reduction and scale/branch families are reviewed too; an independent ternary tower gives continuum many maximal extensions of each fixed dyadic branch, without increasing the total continuum lower bound. The remaining class-residue proofs are reviewed: characteristic-zero maximal quotients separate points and embed in  $\mathbf{No}[i]$ , and binomial specializations fail to preserve Hahn summability. Characteristic-zero support-closed primes correspond to convex subgroups. Term-closedness alone suffices when the recovered subgroup has a set-sized cofinal subset, including principal convex subgroups. The unrestricted closure question and full source reconciliation remain open. Shared support-ring prerequisites, actual surreal scale gaps and the generic scaled-field collision bounds are formalized. The latter bound a module's own cardinality and allow nonunital ring maps. The universal small-target theorem is now proved for both actual omnific rings by the countable telescope: every additive multiplicative map factors uniquely through its ordinary integer or Gaussian integer constant, maps on the purely infinite ideal vanish, and that ideal annihilates every small unital module. Smallness is relative to the birthday universe; targets may be infinite, noncommutative and nonunital. Real nonunital maps correspond to idempotents; Gaussian unital maps correspond to square roots of  $-1$ , without requiring centrality. The arbitrary-coefficient version and large Hahn-subfield instance remain pending. Small quotients of the real omnific ring are now classified as  $\mathbb{Z}$  or  $\mathbb{Z}/n\mathbb{Z}$  for ordinary  $n \geq 1$ ; the characteristic-zero kernel is  $\Pi$ . Gaussian small quotients are their ordinary Gaussian quotients, now proved finite of size  $|d|^2$  for each nonzero Gaussian modulus  $d$ . Prime and maximal ideals with small quotient are also classified in both rings: the purely infinite ideals are prime but not maximal, and all other small prime quotients are the finite residue fields at ordinary prime moduli. Characteristic-zero small images are  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ , with their fraction fields identified separately. If a real omnific ideal misses the purely infinite ideal, explicit ordinal monomials give arbitrarily large small families of distinct residues. Every large quotient has the same unital maps to small rings as its constant quotient. Making a purely infinite element invertible excludes nonzero small unital ring images and modules, including for both actual fields and native localizations. The real principal quotient is small exactly for a nonzero ordinary integer generator. Nonconstant generators give large nonzero quotients; those by  $\omega$ ,  $2+\omega$  and  $1+\omega$  have universal small images  $\mathbb{Z}$ ,  $\mathbb{Z}/2\mathbb{Z}$  and zero, respectively. The finite-congruence closure of every real omnific ideal  $J$  is now proved to be  $J + \Pi$ , also its native topological closure. A principal ideal is dense exactly for unit constant term. Nonconstant irreducibles generate proper dense ideals, and any small-target nonunital map killing such an element is zero. The Gaussian principal-quotient classification is now proved too: exactly nonzero ordinary Gaussian generators give small quotients. The Gaussian purely infinite ideal is an idempotent directed union of real monomial principal ideals with no small generating set, preserved by multiplication by nonzero Gaussian constants. Only the coefficient-general principal-size assertion remains pending. Not refereed, priority not certified. The initial normalization proofs are reviewed: every surreal is almost integral, whereas the integral closure is proper, fine-dense, has zero conductor and needs more than any set of algebra generators. In the countable-support model, exactly  $\text{cf}(\kappa)$  negative monomials generate the ambient field over its integer part, proving the denominator size bound sharp; the normalization's own generator count remains open. Lean now proves uniform algebra absorption and complete integral closure for the actual real and Gaussian fields, also for every intermediate ring, and proves the reciprocal obstruction excluding  $1/2$  from ordinary integral

closure. For both real and Gaussian omnific rings, Lean also proves the strict, dense normalization with zero conductor and no small generating set, in the fine order and surreal-modulus topologies respectively. Both actual normalizations are also proved to have no nonzero finite ring target, even noncommutative, and every proper ideal has an infinite quotient. Lean also proves that a valuation containing the real or Gaussian omnific ring is trivial when its value group is small in the birthday universe; such valuations give value zero to the nonintegral half. Injective monomial substitution and exact binomial coefficients now prove that all actual roots of  $T^m = \omega^g + 1$ ,  $g > 0$ ,  $m \geq 2$ , are integral but outside the real or Gaussian omnific ring, respectively. Their small-target invisibility is now proved as well: every unital small-ring image solves the equation at 1, and every image of an adjoined normalization root has  $m$ th power 1, including in infinite and noncommutative targets. Small-generated real omnific algebras are closed and uniformly discrete, and every Cauchy net in one is eventually constant without a bound on its index or range. The remaining normalization results before the arithmetic fibres are reviewed too. A negative least support exponent prevents integrality, giving an exact test on finite supports. Lean now proves the actual real and Gaussian nonnegative-support and constant slices: their constant coefficient must be an ordinary algebraic integer. The real normalization's rational slice is exactly  $\mathbb{Z}$ . The Gaussian normalization has basis  $1, (\sqrt{3} + i)/2$  over the real normalization  $\mathcal{N}$ , and its additive quotient by  $\mathcal{N}[i]$  is exactly  $\mathcal{N}/2\mathcal{N}$ . The arithmetic-fibre review reaches the finite-partition Boolean-power description over the algebraic integers. An added consequence identifies its rationalization with its total quotient ring; componentwise quasi-inverses prove von Neumann regularity of the rationalization. The review also covers finitely generated ideals and integral Gaussian doubling. Modulo a prime the nilradical is nonzero and idempotent, not finitely generated. The Gaussian product survives modulo 2, while the ordinary quadratic subring reduces to a dual-number algebra mapping onto the diagonal, with a nonzero square-zero kernel. All twenty-two standard arithmetic-fibre results are now reviewed. Each prime ideal of the fibre is determined by a Boolean ultrafilter and a prime of  $\mathbb{Z}$ ; its quotient is  $\mathbb{Z}$  or  $\overline{\mathbb{F}}_p$ . Strict prime chains have at most one inclusion. Under global choice every nonzero element survives in some  $\overline{\mathbb{F}}_p$ , so the fibre has zero Jacobson radical. These added consequences remain pending in Lean. The support-descent and quadratic-branch prerequisites are now reviewed. In the countable-support two-armed model, a positive support gap makes both extended normalization ideals radical, so its unreduced arithmetic fibres already embed in the full fibres. This consequence also awaits Lean. The review now also covers uniform splitting and relative Boolean freeness: a fresh dominant scale supplies independent choices over a smaller reduced fibre, and the full fibre has an atomless proper-class Boolean algebra with no set-sized order-dense core. Finite branch specializations prove these claims without evaluating Laurent series at zero. Scalar Boolean partitions simultaneously split any set of nonzero idempotent matrices. These results remain pending in Lean. All twenty-four standard branching proofs are now reviewed. Every set-indexed family of small-target maps has a common kernel in the fibre containing a nonzero idempotent below every nonzero idempotent. It meets every nonzero ideal and has zero annihilator, although individual branch maps detect every nonzero element under global choice. These consequences also remain pending in Lean. The cyclic-workspace proofs are now reviewed through the finite-place residue criterion and coefficient-span formula. Their nilpotents have exact root-refinement thresholds. Integral Laurent elements can have arbitrary constant coefficient; an explicit additive copy of  $k$  in the local fibre has all products zero and disappears in the full-class fibre. The review now extends through the rational ramification filtration, Puiseux reduction, local valuation structure, graded ring and completion. Refinement kernels are exact filtration levels, every rational level is strict, and each nilpotent eventually dies although no finite stage is reduced. The radical is idempotent, whereas the original extended ideal has a nonzero conormal module. All twenty-seven standard cyclic-workspace proofs are now reviewed, including uniform splitting, continuum Boolean order density and exactly  $2^c$  places above zero. The coefficient fibre has a periodic free resolution with nonzero self-Tor in every degree; the analogous arithmetic complex is not exact. Every coefficient outside the algebraic integers gives an obstruction, expressed by an injective additive  $D$ -module map from  $k/\overline{\mathbb{Z}}_k$  into

the arithmetic annihilator quotient. The countable coefficient fragment has the Cantor clopen Boolean algebra. These claims still await Lean. Lean now also classifies nonunital Gaussian maps to arbitrary small nonunital rings by pairs  $(e, j)$  with  $e^2 = e$ ,  $ej = je = j$  and  $j^2 = -e$ . For both actual omnific rings, the purely infinite ideal is exactly the intersection of all small unital observation kernels. The actual-ring factorizations also hold for targets detected elementwise by small rings, including polynomial rings in arbitrary variable types over small coefficients. Neither ring has a faithful small module; operator representations on any small vector space are classified by integer constants or one operator  $J$  with  $J^2 = -\text{id}$ . Gaussian finite-dimensional conjugacy remains pending in Lean. Lean also identifies both monomial localization images by a lower bound on full support: they contain all ordinary coefficients but are proper subrings and not fields. Quotient-localization reflection now holds for both actual rings, even with arbitrary ideals, large denominator classes and small noncommutative targets. For example, inverting  $\omega + 2$  has the small-target maps of  $\mathbb{Z}[1/2]$ ; the nonzero quotient by  $\omega + 1$  has only trivial small unital modules. The coefficient-general versions remain pending. For both actual rings, a small family of maps with specified nonzero kernel witnesses has a common nonzero positive monomial in every kernel, even with large targets. Lean also proves the real ordered-map and small-retract obstructions; every Gaussian endomorphism induces identity or conjugation on constant terms and preserves and reflects the purely infinite ideal. Four companions add class residue fields, semialgebraic preservers, polyhedral-cone cores and henselian branching (105 standard results). Six henselian foundation proofs now cover the seeded hull, local residue map, survival of every finite branch through localization, Boolean freeness and generic complexification. Six further proofs establish the Boolean-power description, exact branch count and dimension, minimal-prime correspondence, and completion and nilpotent-target behavior. The main theorem is now assembled: a ring of size  $\kappa$  has  $2^\kappa$  minimal primes, yet its completion is only  $\mathbb{Q}$ . Every local map to a separated local  $\mathbb{Q}$ -algebra also factors through  $\mathbb{Q}$ . Formal smoothness is carefully distinguished from the finite-presentation smoothness criterion. The final eight proofs and factor summary complete the review of all twenty-two standard henselian results. Strict henselization adds a Galois factor to the generic spectrum; Gaussian extension doubles it with conjugate complex scalar actions. Finite arithmetic descent also treats non-Galois intermediate fields. Finite projective modules and finite étale algebras are classified by locally constant ranks, including zero; constant-rank projectives are free. The valuation-component count uses the cited external correspondence, without Noetherian assumptions. All twelve question/status notes are now reviewed too. They give a normalization-point/branch correspondence with matching Boolean coordinates, a branch-surjectivity criterion, positive equal dimensions, and perfect-complex object and tensor formulas. At every ordinary prime centre, the local ring and its henselization complete nonflatly to  $\mathbb{Z}_p$ ; forced generic complexification persists even at 2, while a local square root of  $-1$  exists exactly for  $p \equiv 1 \pmod{4}$ . Precise mixed-characteristic branch counts and a prescribed birthday cutoff remain unresolved. These manuscript consequences remain pending in Lean.

## 19. Omnific-preserving automorphisms

[surreal/omnific-preserving-automorphisms/article.pdf](#)

from fifteen manuscripts, integrated through 50dde76

Which automorphisms of a Hahn field preserve its omnific integer part? For  $K = k((t^\Gamma))$ ,  $k$  of characteristic zero and  $\Gamma$  divisible, including  $\Gamma = \mathbf{No}$  and so  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , a strong  $k$ -linear 1-automorphism preserves the omnific ring iff it commutes with the projections onto the purely infinite, constant and infinitesimal parts, iff every monomial is displaced only inside the Archimedean block of its exponent; inclusion already suffices, and necessity needs no strongness. Homogeneous shears and binomial twists realize the stabilizer, which is trivial exactly for Archimedean  $\Gamma$ , contains  $\mathbb{Z} \wr \mathbb{Z}$  over every set of parameters, and has fixed field  $k((t^{B(\Gamma)}))$ ,  $B(\Gamma)$  the bottom convex subgroup. Every field automorphism preserving  $\mathbf{Oz}$  fixes  $\mathbb{R}$  and commutes with the floor. Order, projections and floor are definable in  $(\mathbf{No}, \mathbf{Oz})$ , but with any set of parameters the monomials, the omega-map, simplicity and the Gonshor exponential are not. Part II: root-covered rings such as  $\mathbf{Oz}$  have polynomial coefficient rigidity (an isomorphism  $\mathbf{Oz}[X_1, \dots, X_n] \cong B[Y_1, \dots, Y_n]$  carries

$\mathbf{Oz}$  onto  $B$ ), no nontrivial gradings or torus coactions, and formal flows that no polynomial family realizes. *One classification proved three times, printed once from the most general source; a fifth manuscript adds the support-cut form. The proper-class correspondence between derivations and automorphisms is imported, and the statements depending on it are marked.* Part III, added in 20c4c9f, claims automatic strongness of every  $\mathbf{Oz}$ -automorphism and proper strong embeddings moving real coefficients. Strongness for arbitrary  $\mathbf{Oz}[i]$ -automorphisms without conjugation or valuation compatibility remains open in the report. Source 14 adds automatic summability criteria and formal-power-series covariance in Part IV. Sources 15–16 add topological collapse and arbitrary-rank coefficient recovery. Sources 17–19 add formal orbit fields, actions of set-sized left-orderable groups with prescribed fixed fields, and formal integration of omnific derivations. The surcomplex group classification names conjugation; the formal flows use an indeterminate rather than an evaluated infinite series. These new claims await proof review and Lean formalization; not refereed, priority not certified. Source 20 adds universal symmetries and exact difference equations; its 24 new standard results also await independent review and Lean formalization.

## 20. Independent surreal copies and arithmetic intersections

[surreal/independent-surreal-copies/article.pdf](#)

from three manuscripts, integrated through 32bb723

Proposed strong self-embeddings with prescribed common full Hahn field on a set-sized divisible exponent subgroup, finite-family linear disjointness, and preservation and reflection of omnific integers. The report distinguishes intersection of fraction fields from the fraction field of intersected rings, and a full Hahn field from an ordinary compositum, with claimed transcendental gaps. Its class construction uses NGB with Global Choice. The source and PDF are indexed and built; mathematical proof review, source reconciliation and Lean formalization remain pending. The second manuscript adds exact rank tests, differential transcendence and omnific witnesses beyond ordinary composita. The third treats maximal transcendence in Hahn joins and prime omnific witnesses. Their 52 new standard results await independent review and Lean formalization.

## 21. Omnific groups and lattices

[surreal/omnific-groups-and-lattices/article.pdf](#)

from five manuscripts, delivered 23 September 2026

Which algebraic groups and lattices gain omnific points? A real algebraic group has no new points over  $\mathbb{R} \oplus \Pi$  iff its identity component is reductive with compact real derived group, and for a closed subgroup scheme of  $\mathrm{GL}_{N,\mathbb{Z}}$  this is  $\mathcal{G}(\mathbf{Oz}) = \mathcal{G}(\mathbb{Z})$ ; over  $\mathbb{C}$  the condition is that the identity component be a torus. Otherwise  $(\Pi, +)$  embeds in the constant-term kernel, so  $\mathrm{SO}_3(\mathbf{Oz})$  has 24 elements while  $\mathrm{SO}_3(\mathbf{Oz}[i])$  has unipotent families. For  $n \geq 3$  the constant-term kernel of  $E_n(\mathbf{Oz})$  is generated by purely infinite root elements and is perfect, torsion-free and centerless with no nontrivial set-sized image, so  $\mathrm{SL}_n(\mathbb{Z})$  is the universal set-sized quotient of  $E_n(\mathbf{Oz})$ , and with Bass–Milnor–Serre its profinite completion is  $\mathrm{SL}_n(\hat{\mathbb{Z}})$ ; Steinberg versions and exact thresholds at set-sized stages follow. In rank two the kernel is a free product of copies of  $(\Pi, +)$  and no universal set-sized quotient exists. Part III: in an omnific lattice with a separated positive energy a shortest vector exists iff an explicit integral kernel is nonzero; otherwise the energies have no surreal infimum, and Gauss and LLL reduction can fail. *Merged from five manuscripts; the theorem for  $n \geq 3$  is proved by different routes, and both are kept.* Not claimed:  $\mathrm{SL}_n(\mathbf{Oz})$  in place of  $E_n(\mathbf{Oz})$ ,  $K_2$ , a surreal volume theory. Proof review pending; not refereed, no Lean formalization, priority not certified.

## 22. Discrete initial subgroups and omnific normalization

[surreal/discrete-initial-subgroups-and-omnific-normalization/article.pdf](#)

from two manuscripts, delivered 23 September 2026

Is every discretely ordered initial subgroup of  $\mathbf{No}$  isomorphic to an initial subgroup of the omnific integers? This question of Ehrlich and Kaplan (arXiv v1 Question 2; *J. Symbolic Logic* 2018, Question 9.1) receives a proposed affirmative answer. The least positive element is  $2^{-n}\omega^{-\alpha}$ ; an explicit relocation of the sign-tree root and a real-linear transport of Hahn spaces carry the group onto an initial  $H \subseteq \mathbf{Oz}$  with that element sent to 1. The possible images are exactly the initial subgroups of  $\mathbf{Oz}$  containing an explicit dyadic-spine group; the bottom cyclic layer splits off canonically, and Gaussian-linear surcomplex versions hold. The second manuscript proves every convex subquotient of an initially realizable group initially realizable, answering the report's questions on convex quotients and limit stages. For Presburger groups, realizability is equivalent to the least positive element generating a direct summand;  $2^{\aleph_0}$  rigid countable  $\mathbb{Z}$ -groups are not realizable, realizability is not elementary, and no nonstandard model of Peano arithmetic has a realizable additive group. *Each manuscript's proofs were checked by hand when placed; that is not refereeing.* The Ehrlich–Kaplan initiality criterion is imported, not re-proved; the transport is neither a simplicity isomorphism nor multiplicative; the literature search was targeted; no other Ehrlich–Kaplan question is answered. Not refereed, no Lean formalization, priority not certified.

## 23. Exact precision and algebraic rigidity of omnific continued fractions

[surreal/omnific-continued-fractions/article.pdf](#)

from one manuscript, placed in aa9c891, 23 September 2026

Which surreal numbers have the same infinite sequence of omnific continued-fraction digits? The report describes each fiber as a translate of the valuation ideal  $v(e) > 2 \deg(q_n)$  for every ordinary finite  $n$ , where  $q_n$  is the continuant denominator. It gives realization and uniqueness criteria in full Hahn fields, an algebraic-fiber classification over the digit field, and a distinguished realization of degree at most two for eventually periodic codes. Full surreal fibers contain independent families of every set cardinality over a prescribed set-sized coefficient field. The surcomplex construction is coordinatewise. Roggeman's prior nonuniqueness result is credited; there is no transfinite continued-fraction algorithm or general Lagrange converse. Independent proof review and Lean formalization remain pending.

## 2 The surcomplex plane

Twenty-eight reports about  $\mathbf{No}[i] = \mathbf{No}[i]$ . Seven develop a common holomorphic-function framework. The others study differential equations, dynamics, support obstructions, entire functions, spectral theory, theta functions, positivity, reconstruction, summability, duality and automorphisms. Later additions extend several of these reports; their provenance sections identify the integrated manuscripts and the work still awaiting assembly. The autonomous-dilation base placed in 21375f8 is now written in 260c143; its two companions are integrated in 1ef6bba, adding support-rank criteria and relative independence of dilation orbits.

*Read analysis first.* It is the foundation the other six continue. In the *full fine topology*, every set-sized subset is closed and discrete and every convergent set-indexed net is eventually constant. Universe-relative formulations require smaller-universe subsets and indices. These conclusions do not apply to the intrinsic valuation topology of a fixed Hahn field:  $t^n \rightarrow 0$  in  $\mathbb{C}((t^{\mathbb{Q}}))$ . Strong Hahn summability is a separate support condition; even in rank one it need not give convergence of finite subsums, as the foundations report's  $\sum t^{n/(n+1)}$  demonstrates. Contour integration in the analytic reports is defined coefficientwise on ordinary complex paths.

*Then read the ring table in analytic-geometry, before anything else.* Four coefficient rings circulate through this material under nearly identical notation — the radius-free  $\mathbb{C}\{z\}((t^{\Gamma}))$ , the fixed-polydisk  $\mathcal{O}(D)((t^{\Gamma}))$ , the common-domain germ ring, and the formal  $\mathbb{C}[[z]]((t^{\Gamma}))$ . These are

not an interchangeable chain. For  $\eta > 0$ , the family  $F(z) = \sum_{r \geq 1} t^{r\eta}/(1-rz)$  lies in the radius-free germ ring but not the common-domain germ ring: its coefficient poles  $1/r$  approach zero. This witnesses that particular strict inclusion. The report gives separate fixed-domain obstructions and explains which local proofs require shrinking. The remaining five reports can be read in any order.

## 24. Surcomplex analysis: holomorphic functions on $\mathbf{No}[i]$

[surcomplex/analysis/article.pdf](#)

from `surcomplex_analysis.zip` and eight companion archives

One article assembled from nine manuscripts delivered on 21 September 2026. It develops Hahn-summed local germs, coherent sections over connected ordinary complex domains, preparation and finite zero clusters, contour residues, exponentials and logarithms, and polynomial or rational rigidity under coherence at every positive surreal scale. Each formal series has a sufficiently small realization, depending on the series; evaluation is faithful as a function germ, not at one fixed point. The uniform Taylor-remainder bound gives the full fine derivative. Canonical lifts, coherent sections and germ-analytic functions form strictly increasing classes; germ analyticity implies fine differentiability, with no converse or strictness theorem for that last inclusion here. Coherence supplies global identity, maximum and open-mapping principles over the ordinary base. Sharp Schwarz–Pick bounds require canonical lifts. An ordinary-real value bound and coefficientwise boundedness are incomparable:  $tZ$  satisfies the first but not the second, while the constant  $\omega$  satisfies the second but not the first. Only the coefficientwise Liouville hypothesis forces constancy; one arbitrary surreal bound is always available. For  $1/(\zeta - \varepsilon)$  with nonzero infinitesimal  $\varepsilon$ , the cluster residue at the ordinary centre 0 is 1, while the actual local residue at 0 is 0. Rouché’s halo theorem requires an ordinary margin; a bare strict surreal boundary inequality can miss a zero in a boundary monad. The global and cut-plane logarithms differ by  $2\pi i$  on specified points. Every explicit exponential twist retains enough cofinal periods to give the proved value distribution of  $E_\lambda(1/z)$ , although their kernels need not be nested; this is not a general Picard theorem. Full fine topology and the intrinsic topology of a fixed Hahn workspace remain distinct. The nine finite verification programs and historical `MERGE_NOTES.md` are preserved; the maintained `README` identifies corrected comparisons in that older record.

## 25. Infinitesimal analytic geometry over the surcomplex numbers

[surcomplex/analytic-geometry/article.pdf](#)

from `Surcomplex_Analytic_Geometry.zip` + `surcomplex_finite_geometry.zip`

The ring theory of the infinitesimal neighbourhood — the monad  $\mathfrak{m}_K^n$  of an ordinary point — rather than the deformation theory of the reports that follow. Proves, *separately of each of* the radius-free  $A_n = \mathbb{C}\{z_1, \dots, z_n\}((t^\Gamma))$  and the common-domain  $\mathcal{R}_n$ , that it is a Noetherian Jacobson domain of dimension  $n$  whose maximal ideals are exactly the infinitesimal evaluation ideals  $\mathfrak{m}_a = (z_1 - a_1, \dots, z_n - a_n)$  with residue field  $K$ , regular local at each, with completion  $K[[X_1, \dots, X_n]]$  — and that neither is local, each recording every infinitesimal point at once. Carries Weierstrass division and preparation, a weak and a strong Nullstellensatz, Noether normalization, finite projection, contour residue duality, Hensel–Rouché, and workspace invariance: every common zero over any larger divisible workspace already lies in  $K_\Gamma^n$  with unchanged multiplicity, so the count accounts for all zeros in the whole proper class  $\mathbf{No}[i]$ . The nonzero-value-group hypothesis is essential and not aesthetic — for  $\Gamma = \{0\}$  the strong Nullstellensatz fails at  $I = (0)$ . *This is the report that fixes the four-ring vocabulary and proves the inclusions strict; statements holding over both  $A_n$  and  $\mathcal{R}_n$  are stated twice with both proofs, since the two proofs use different resources and neither transfers.*

## 26. Finite Hahn deformations: division, conservation of multiplicity, residue duality

[surcomplex/finite-deformations/article.pdf](#)

from `surcomplex_finite_deformations.zip`, `_finite_fibres`, `_finite_maps`, `_intersections`, `_research`

Five manuscripts on one theorem: an ordinary holomorphic complete intersection  $f$  on a fixed polydisk, with the origin as its only common zero and colength  $\mu$ , keeps its entire finite geometry under any Hahn perturbation  $F = f + E$  of well-ordered *strictly positive* error support — on the unshrunk domain, with an explicit support certificate  $\text{supp} \subseteq \text{supp}(G) + M$ ,  $M$  the monoid generated by the error support. The remainder of division is unique; the witnesses need not be, their ambiguity being a Koszul syzygy carried by the same certificate. Gives the integral structure  $H_{\Gamma}^+(D)/(F) \cong (K^\circ)^\mu$  over a valuation ring that may have high-rank value group and need not be Noetherian, with nothing completed; conservation of multiplicity at the *actual* displaced surcomplex zeros rather than their standard parts,  $\sum_{\zeta} \mu_{\zeta} = \mu$ ; and a coefficientwise residue  $\text{Res}_F$  — defined by a series, not by a surcomplex contour integral — whose Gram determinant is a unit even at a collision and which satisfies  $\text{Res}_F(G \cdot J_F) = \text{Tr}(M_G)$ . The separate residue formula, when all zeros are simple, is  $\text{Res}_F(G) = \sum_{\zeta} G(\zeta)/J_F(\zeta)$ ; for nonnegative-support  $G$ , individual terms can be infinite while the total is finite. Also Hartogs, characters and local algebras, spectral invariants, trace-Jacobian and Bézoutian, monodromy and stability. For  $\mu = 1$ , the inverse-map proof constructs one uniform Hahn support before evaluation. The engine throughout is positive-support Neumann inversion; ordinary analytic convergence occurs in finite-parameter comparisons, while the Hahn constructions use strong support summation.

## 27. Jordan separation, Cauchy theory, and Stokes calculus on the surcomplex plane

[surcomplex/contours-and-stokes/article.pdf](#)

from `surcomplex_contours.zip` and two same-named archives

Three manuscripts whose titles differ by one word — Stokes *Calculus*, *Integration*, *Formulae* — over one body of work. All three refuse to transfer the Jordan–Cauchy–Stokes package to  $\mathbf{No}[i]$  as a unit, and all three split it the same way: a coarse Jordan separation theorem in the standard-part topology whose separator is the whole boundary monad rather than a curve; an imported semialgebraic Jordan theorem; a Hahn-coefficient de Rham complex with coefficientwise Stokes, extended to infinitesimally deformed chains; and the holomorphic consequences — Cauchy, endpoint displacement, integer winding, and microscopic circle and torus realizations of the perturbation residue — with an algebraic rational contour pairing characterized uniquely by vanishing on exact differentials and normalizing  $dz/(z - a)$  to  $2\pi i \cdot \text{wind}_{\text{sa}}$ . Adds a purely algebraic third chain category over any characteristic-zero field, a support-preserving Poincaré lemma, an order-valued ML estimate, and a root-of-unity/Schirelman section. This is where the question the deformation report leaves open is answered: supplying an actual surcomplex contour that represents the residue series. *Two thirds of each manuscript was the same mathematics in the same order; 82 numbered results became 59, by duplicate elimination only.*

## 28. Global support obstructions: moving divisors, Mittag–Leffler, and a Picard dichotomy

[surcomplex/global-divisors/article.pdf](#)

from `surcomplex_global_support`, `_global_theorems`, `_compact_curves`

The local-to-global theory: when do infinitely many individually legitimate infinitesimal prescriptions admit one coherent global realization over the ordinary complex plane? The sharp answer for divisors is a *symmetric-support* criterion — a cluster family  $P_c(u) = u^{m_c} + \sum_{j < m_c} a_{c,j} u^j$  over a closed discrete ordinary  $A \subset \mathbb{C}$  is the exact zero divisor of a nonzero global section iff  $\bigcup_c \bigcup_j \text{supp}(a_{c,j})$  is well ordered: the supports of the symmetric coefficients, not of separately chosen roots. Hence the two discriminating examples —  $n + t^{1/n}$  alone is not realizable, its descending symmetric support  $\{1/n\}$  failing, while *all* roots of  $(z - n)^n = t$  are, symmetric support the singleton  $\{1\}$  — and hence a globally principal effective divisor with a nonprincipal effective sub-divisor. Also Mittag–Leffler at fixed centres and for moving poles, a support-restricted Chinese remainder theory with an explicit Neumann right inverse, an exact Cousin criterion in terms of

singular support, and the Picard dichotomy: every ordinary-base stalk is a principal ideal domain, nonlocal when  $\Gamma \neq 0$ ;  $\text{Pic}(\mathbb{C}, \mathcal{H}_\Gamma) = 0$  iff  $\Gamma = \{0\}$  or  $\Gamma = \mathbb{Z}\delta$ , and otherwise  $\dim_{\mathbb{C}} \text{Pic} \geq 2^{\aleph_0}$ , with equality for countable noncyclic  $\Gamma$ . A third manuscript studies a compact connected ordinary Riemann surface  $X$  of genus  $g$ . Its Picard group splits into  $H^1(X, \underline{\Gamma}) \oplus \text{Pic}(X) \oplus (H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \mathfrak{m}_\Gamma)$ . The valuation class is exactly the obstruction to a nonzero coherent meromorphic section, even in arbitrarily high degree. In the zero-valuation sector a finite matrix with positive-support entries computes cohomology and gives Riemann–Roch. A root-free Newton-sum Abel functional decides principality of finite signed clusters. The one-scale comparison has kernel  $H^1(X, \mathcal{O}_X) \otimes_{\mathbb{C}} \bigcap_n t^{n\delta} \mathcal{O}_\Gamma$ ; it is injective exactly when  $g = 0$  or  $\delta$  is cofinal in  $\Gamma$ . Any prescribed set of surreal cutoffs can be defeated after enlarging the workspace; all positive cutoffs detect classes in a fixed nonzero workspace. Coordinate charts transfer the local PID argument to  $X$ , but the global plane and compact classification theorems retain their different base hypotheses.

## 29. Surcomplex polynomial algebra: root geometry, Newton profiles, residue duality

[surcomplex/polynomial-algebra/article.pdf](#)

from `surcomplex_polynomial_algebra.zip` and two same-named archives

Finite-degree polynomial algebra over  $\mathbf{No}[i]$ , worked inside set-sized algebraically closed Hahn workspaces and keeping two geometries deliberately apart throughout: the surreal-valued modulus, whose disks are ordered, and the natural Hahn valuation, whose balls have arbitrary ordered-group rank. A Rouché theorem in each. Covers the finite polynomial algebra and the fundamental theorem there, CRT/Hermite jets, resultants and discriminants, ordered root geometry (root bounds, Gauss–Lucas, Jensen disks, a sharp Schoenberg second-moment inequality with its equality case), real-closed-field transfer with Hermite signature counting, exact root loci under coefficient uncertainty in both geometries, Newton profiles over an arbitrary ordered exponent group, image balls and branch values, exact Rolle counts and the finite root-cluster tree, support-controlled coprime Hensel factorization with a fixed-domain parameter form that shrinks no neighbourhood, and universal residue duality with the trace identity. *Three perturbation theorems with three different hypotheses are kept separate rather than collapsed, as are two discriminant layer formulas and two mapping theorems, with their agreement proved; the alternative was to lose what each source reached that the others did not.*

## 30. Trigonometry on the surcomplex plane

[surcomplex/trigonometry/article.pdf](#)

from `surcomplex_trigonometry.zip` and two same-named archives

Its organizing theorem is that every surcomplex direction has a *finite* surreal angle:  $\text{cis} : (\mathcal{O}_{\mathbb{R}}, +) \rightarrow (T(\mathbf{No}), \cdot)$  is surjective with kernel exactly  $2\pi\mathbb{Z}$ , so  $\mathcal{O}_{\mathbb{R}}/2\pi\mathbb{Z} \cong T(\mathbf{No}) \cong T(\mathbb{R}) \times (\mathfrak{m}_{\mathbb{R}}, +)$ , and the whole of plane, spherical and hyperbolic triangle geometry is available at arbitrary surreal length scale without ever choosing a value for  $\sin \omega$ . From it: canonical sine and cosine by Hahn summation, the unit circle with its branch conventions and projective half-angle chart, inverse functions on their full surreal domains, the angle-sum, cosine, sine, tangent, half-angle, Heron, incircle, Euler, angle-bisector, cevian, Ceva, inscribed-angle and Ptolemy theorems, an exact valuation dictionary between sides, angles and area with a quadratic triangle-inequality defect formula and two flat-triangle theorems, a rank-two collision-stable intersection algebra, the canonical complex extension to the strip where sine is surjective with completely determined fibres, finite Fourier inversion, Parseval and a surcomplex Fejér–Riesz factorization, and the classification of all global phase extensions, with the Ehrlich–Kaplan normalization placed inside that classification. Two coherence obstructions close it: there is no coherent bounded sine of infinite frequency, and all-scale coherence forces a polynomial. *No mathematical disagreement was found between the three sources; the duplication was near-total on the first two thirds and the divergence near-total on the last.* A later manuscript adds Section 18 on  $\text{SO}(2, \mathbf{No}) \cong \mathcal{O}/2\pi\mathbb{Z} \cong \text{SO}(2, \mathbb{R}) \times (\mathfrak{m}, +)$ : isometries are automatically linear, a rational projective chart, torsion  $\mathbb{Q}/\mathbb{Z}$ , one real coordinate per valuation layer, a strongly summable logarithm, non-algebraic group automorphisms and the fine topology; what it shares with the circle-group results above is printed once. Two further manuscripts

(Section 19) study the periods of global phases: their profinite defect, and normalized surcomplex exponentials whose kernel-multiplier ring is exactly  $\mathbb{Z}$  without being an integer part, which refutes one sufficiency possibility in the first Ehrlich–Kaplan robustness question; the question itself stays open.

### 31. Differential algebra and differential equations over the surreal and surcomplex numbers

[surcomplex/differential-equations/article.pdf](#)

from eleven manuscripts, delivered 21 and 22 September 2026

Eleven manuscripts on the Berarducci–Mantova derivation  $\partial$  on  $\mathbf{No}$ , normalized by  $\partial\omega = 1$ , and its unique extension to  $\mathbf{No}[i]$  — unique because differentiating  $i^2 + 1 = 0$  forces  $\partial i = 0$ . Seven share a twelve-item spine, printed once; four later members each add a part. The organizing identity is exact: the logarithmic-derivative image is  $\mathbf{No} + i\mathcal{A}$ ,  $\mathcal{A} = \partial\mathfrak{m} = \partial\mathcal{O}$ , the two forms agreeing only through a proved lemma, so  $\partial y = (a + ib)y$  has a nonzero surcomplex solution exactly when  $b$  has a *finite* primitive.  $\mathcal{A}$  is a *strict* subclass of  $\mathfrak{m}$ : the infinitesimal  $\omega^{-1}$  has the infinite primitive  $\log \omega$ , so  $\partial y = i\omega^{-1}y$  has only  $y = 0$  — collapsing  $\partial\mathcal{O}$  to  $\mathcal{O}$  would make it solvable, which the first eight sources all refute. Hence no formal oscillator:  $\partial y = iy$  and  $\partial^2 y + y = 0$  have only  $y = 0$ . Also the purely infinite part of the accumulated phase as a complete gauge invariant in a four-term exact sequence; the sharp threshold  $\text{minsupp } b > 1$  over real-exponent Hahn fields, with a counterexample pair against globalizing it by valuation; Picard–Vessiot extensions with diagonal phase tori; and a separate coordinate half with a coherent nonlinear initial-value theorem. Part III (member 10): every  $\partial y = Ay$ ,  $A \in \text{Mat}_n(\mathbf{No}[i])$ , is gauge equivalent to  $i \text{diag}(\partial P_j)$ , the multiset  $\text{Ph}(A) = \{P_j\}$  of purely infinite surreals being a complete invariant (Theorem 15.9), computed for Hermitian coefficients from the algebraic eigenvalues with no commutation or gap assumed (Theorem 16.5). Example 16.10 refutes extending that eigenvalue formula to arbitrary non-Hermitian coefficients; the abstract classification still covers them. The normal form and inhomogeneous existence rest on an imported splitting/surjectivity theorem. Part V (member 12):  $\partial y = -tAy$ ,  $t = \omega^{-1}$ , with ordinary complex residual matrix  $A_0$  and positive-order power-Hahn perturbation, classified over  $\mathbb{C}((t^{\mathbb{R}}))$  and over  $\mathbf{No}[i]$ ; the solution space has dimension the algebraic multiplicity of the real eigenvalues of  $A_0$ , and, with  $\lambda_j$  the eigenvalues of  $A_0$ ,  $\text{Ph}(-tA) = \{(\text{Im } \lambda_j) \log t\}$  is a non-Hermitian sector reached without that import. Part VI (member 11): first-order autonomous equations with constant coefficients on curves, classified (Theorem 30.1) with the same solutions for *every* normalized derivation (Theorem 31.4), no two asserted equal on  $\mathbf{No}[i]$ , and the logarithmic-derivative image of a constant abelian variety. *Merged; the hazard was a word.* “Phase” named the finite angle in  $\mathcal{O}$ , the infinitesimal residue in  $\mathfrak{m}$ , and the accumulated phase, no angle at all; one source printed two of them twenty lines apart. Each gets one symbol,  $\text{Ph}(A)$  collects obstructions of the third, member 12’s “phase labels” become residual frequencies,  $J$  is retired, and the meaningless merged sentence is printed beside the correct one. Later reproofs are cited as agreements; hedges that later members answer, some only in part, are amended in place, wording kept. Section 51 registers 106 non-claim items: the derivation is imported, not reproved, and no priority, refereeing or proof-assistant verification is claimed. Corollary 11.4 strengthens in phase-independence the prior chosen-exponential obstruction of [surcomplex/analysis](#);  $\text{Ph}$  is not the spectrum of [surcomplex/spectral-theory](#). Part VII (member 13): on the log-atomic tower,  $\partial^2 y + (Q_\alpha + c(\ell'_\alpha/\ell_\alpha)^2)y = 0$  has a two-dimensional solution space for  $c \leq 1/4$  and none for  $c > 1/4$  at every ordinal depth (Theorem 38.1); no set of earlier scales decides this (Theorem 39.2); at limit stages three nested Hahn fields give dimensions 0, 1, 2, with a Borel Galois group (Theorems 40.4, 41.2).

### 32. Rank-one non-Archimedean analytic geometry inside the surcomplex numbers

[surcomplex/rank-one-berkovich/article.pdf](#)

from [surcomplex\\_rank\\_one\\_berkovich.zip](#)

The report devoted to intrinsic rank-one nonarchimedean analytic convergence. Inside the rank-one workspace  $K = \mathbb{C}((t^{\mathbb{R}}))$  the absolute value  $|a|_v = \exp(-v(a))$  is genuine and  $K$  is spherically complete, complete and algebraically closed, so it is a legitimate base for Tate algebras and Berkovich spaces. Proves strict inclusions among the Tate, polynomial-coefficient Hahn, common-domain, radius-free germ, formal-coefficient and open-disk analytic rings with named witnesses, extending the ring table of [surcomplex/analytic-geometry](#); gives a constructive one-variable preparation theorem, principal-ideal and finite-quotient results, analytic root counts and a valuation Rouché theorem; constructs the Berkovich disk points and the annulus retraction; and contrasts the contractible annulus spectrum with its nonzero analytic de Rham class. The worked case study is the entire series  $F(Z) = \sum_{n \geq 0} t^{n^2} Z^n$ , taken from [foundations-and-computation/foundations](#) and credited there: its full zero geometry is settled, and the first correction of the  $m$ th normalized root occurs at  $t^{m(m+1)}$ . *This does not contradict the no-convergence theorems of [surcomplex/analysis](#).* The convergence here is in the intrinsic valuation topology of one fixed rank-one field, not the fine topology induced from  $\mathbf{No}[i]$ ; and at  $Z = t^{-\omega}$  the exponents  $n^2 - n\omega$  strictly decrease, so the support is not well ordered and  $F$  is not an all-surcomplex entire function — the obstruction agrees with all-scale rigidity. The original audit credits the Berkovich *comparison* in [surcomplex/contours-and-stokes](#); its development-gap claim concerns that inspected snapshot, not an exhaustive search of the current collection.

### 33. Surcomplex spectral theory: singular values, infinitesimal rank, multiscale stability

[surcomplex/spectral-theory/article.pdf](#)

from three manuscripts, delivered 21 and 22 September 2026

Finite matrices over a real closed field  $F$  and  $F(i)$ , then over set-sized Hahn workspaces  $K_{\Gamma} = \mathbb{C}((t^{\Gamma}))$  inside  $\mathbf{No}[i]$ ; every dimension is an ordinary finite integer, and the analytic coefficient-ring machinery is not needed. The base manuscript proves the spectral theorem and the SVD *without compactness* (classical, not claimed as new): real closedness suffices, the witnesses are exhibited, and Courant–Fischer and Ky Fan extrema are *attained*; then positive square roots, polar decomposition, generalized Hermitian eigenproblems, Moore–Penrose inverses, least squares, best low-rank approximation and conditioning. Its core identifies singular-value valuations with successive differences of minimum minor valuations  $\Delta_k(A)$ , and recovers them and the leading singular coefficients from positive Gram coefficients one scale block at a time; applications include an intrinsic rank filtration, root-free scale elimination, precision certificates, nonnormal pseudospectra, gap-sensitive projectors and Schur-complement corrections. The second manuscript drops divisibility of  $\Gamma$ : Hermitian and normal descent and the SVD keep every exponent in the subgroup generated by the input supports, through a finite splitting tree; for positive semidefinite  $A$  and integer  $q \geq 2$  the principal  $q$ th root, and equally its trace, generates the field with exponent group  $\Gamma + \sum_{k=1}^{\text{rank } A} \mathbb{Z} \Delta_k(A)/q$ , of degree the order of the subgroup the  $\Delta_k(A)$  generate modulo  $q\Gamma$ ; at  $q = 2$ ,  $\Delta_k(A) \in 2\Gamma$  for  $1 \leq k \leq \text{rank } A$  is the exact Gram-factorization criterion. The third (Section 12) treats monic squarefree real-rooted  $p$  over  $F_{\Gamma} = \mathbb{R}((t^{\Gamma}))$ ,  $\Gamma$  arbitrary: with  $h_1, \dots, h_n$  the leading principal minors of its Newton-sum Hankel matrix and  $h_0 = 1$ , the splitting field is exactly  $F_{\Gamma}(\sqrt{h_1}, \dots, \sqrt{h_n})$ ,  $p$  has  $\#\{k : v(h_k) - v(h_{k-1}) \in 2\Gamma\}$  irreducible factors, and both persist for every monic  $\tilde{p}$  of degree  $n$  over  $F_{\Gamma}$  with  $v(h_k(\tilde{p}) - h_k(p)) > v(h_k(p))$  for all  $k$ ; applied to the squarefree part of  $\det(XB - A)/\det B$ , this gives the exact field of the generalized eigenvalues of a definite Hermitian pencil, not of a normalized eigenbasis. The pivot-class profile does not determine the factor degrees. Shared results are printed once, in the earlier manuscripts’ form; the third’s Hensel-lifting proof of real principal axes is kept as a marked second route, and its section sits here rather than in [surcomplex/polynomial-algebra](#), whose divisible workspaces make every such invariant vanish. Priority is not certified; not refereed or proof-assistant verified; the three repository audits were targeted and establish no absence

elsewhere. `surcomplex/prony-reconstruction-at-surreal-scales` shares only the classical Hankel moment identity; `surcomplex/infinite-dimensional-hahn-spectral-theory` is the infinite-dimensional sibling, and no theorem of it is used here.

### 34. Surcomplex dynamics: linearization thresholds, periods, and normal forms

`surcomplex/dynamics-and-normal-forms/article.pdf`

from eleven manuscripts, integrated through 16fc51c

Local and global dynamics over complex Hahn fields  $K = \mathbb{C}((t^\Gamma))$  realized inside  $\mathbf{No}[i]$ , with  $\Gamma$  a set-sized ordered abelian group of *arbitrary* valuation rank. The first ten manuscripts combine nine treatments of germ and fixed-point dynamics with one (§§17–19) on the quasi-periodic half, a cohomological equation on a real torus whose frequency vector has *surreal* components, with divisors  $k \cdot \Upsilon$  indexed by an integer lattice. **Read Section 2 first:** five inequivalent conditions travelled as *resonance*, six as *small divisor*, and two groups of sources used *radius* with **opposite** monotonicity; each variant is named, every threshold states its variant and coefficient category, and Section 2.5 is a tag ledger. Lattice resonance is *not* frequency resonance, and visible divisor valuations are *not* a spectrum. Quasi-periodic results:  $d$  surreal components give *at most*  $d$  arithmetically visible divisor valuations, via a canonical descending flag of saturated integer resonance lattices of strictly dropping rank, with one well-ordered set containing every reciprocal nonzero divisor’s support (not a summability claim); an exact stratified subexponential criterion for universal solvability on one fixed torus strip; an arbitrary-order lifting obstruction; and, with no lattice resonance, under that criterion and for perturbations of valuation above the loss  $\kappa_\Upsilon$ , a unique normalized mean-zero conjugacy to a constant vector field with a frequency correction, by marked support words rather than sequential valuation convergence. Germ results: the exact maximal centered valuation ball read off the finite  $q$ th return iterate; a five-row coefficient-category table, necessity witnessed per row at one exponent; an algebraicity dichotomy over  $\mathbb{C}(w)((t^\Gamma))$  with monodromy and finite-cover obstruction; periods as a complete conjugacy invariant with explicit moduli; a gauge-normalized symplectic normal form with no arithmetic bound. Sources 08 and 09 (§7; source 08 also §8.6) answer the common-domain question for an *exact scalar* multiplier, oppositely to source 02’s drifted answer. For  $\lambda I_d$ ,  $\lambda = e^{2\pi i\theta}$  with  $\theta$  irrational, any nonzero  $\Gamma$ ,  $d \geq 1$ ,  $\tau < \infty$  and input coefficients  $f_s$  holomorphic on one polydisk of radius  $R$  with  $f_s(0) = 0$ ,  $Df_s(0) = 0$ , all coefficients of the normalized conjugacy and its inverse are holomorphic on the open polydisk of optimal universal radius  $Re^{-\tau}$  (Theorem 7.13, credited to both). Under *drift*, beyond source 02’s  $\text{rad}(h_{k,1}) = R \exp(-(k+1)\sigma)$  and its trichotomy, source 08 shows that one Hahn monomial over  $\mathbb{Z}$  already gives radii exactly  $R \exp(-m\sigma)$  for  $0 < \sigma < \infty$  (Theorem 8.15); in one variable over any fixed nonzero  $\Gamma$ , every input’s normalized linearizer has a common positive coefficient disk iff  $\sigma = 0$  (Corollary 8.17). Source 08 is the base (every  $d$ , and the drifted theorem); source 09, translated from its forward convention (its  $H$  is the report’s  $H^{-1}$ ), adds the tree product bound (Theorem 7.5) and a plane-tree proof for  $d = 1$ , kept as a marked second route sharing the arithmetic lemmas. Source 10 (§§7.7–7.13) settles the nonscalar exact case: every exact nonresonant diagonal unitary  $\Lambda$  with  $\tau < \infty$  has a common polydisk of radius  $Re^{-r\tau}$ ,  $r$  the rational rank of the angles (Theorem 7.23), sharp when  $r = 1$ ; the optimal radius for  $r \geq 2$  stays open (Question 25.1). Section 26 records the earlier sources’ limitations. Extends `surcomplex/differential-equations`, but uses external coordinate derivatives, not its Berarducci–Mantova derivation; kept apart from the disks of `surcomplex/rank-one-berkovich`. Not refereed; priority not certified; Lean checks only the support lemma, the combinatorial ancestry bounds of Corollary 3.5 (corrected to keep the input support) and the counterexample to an input-independent bound, not that corollary’s operator-summability assertion. Source 11 adds rotation-equivariant maps and a threshold extracted from the map itself; these 18 new standard results await independent proof review and Lean formalization.

### 35. Nonabelian support obstructions: a matrix Cousin criterion and the surcomplex Riemann–Hilbert problem

[surcomplex/nonabelian-support/article.pdf](#)

from three manuscripts, two delivered 21 September 2026 and one 22 September 2026

One coefficient category, studied from two sides and then past its integrality boundary: matrices  $I_r + \text{Mat}_r(J_\Gamma)$ , congruent to the identity modulo positive Hahn exponents over a set-sized, possibly non-Archimedean value group  $\Gamma$ , with ordinary holomorphic coefficients on a *single fixed* ordinary complex domain ( $t^\gamma = \omega^{-\gamma}$  reads them in  $\mathbf{No}[i]$  when  $\Gamma \subseteq \mathbf{No}$ ). Part I solves a matrix Cousin problem on a puncture-and-disk cover of a connected open Riemann surface, with integral vector bundles and a Stein branch; Part II inverts monodromy for near-identity representations of  $\pi_1$  of a plane minus a closed discrete set, conversely to the forward map of [surcomplex/differential-equations](#). **Both headline theorems read “the union of the supports is well ordered” and are *not* the same criterion:** gluing (Theorem 6.2) tests the *normalized polar factors*  $\text{Pol}(G_a) - I_r$ , each transition factored independently and first, and well-ordering of full raw transition support is sufficient but not necessary, whereas raw singular-coefficient support fails in *both* directions. Realizability (Theorem 14.2) tests the *raw* monodromy matrices  $\rho(\ell_d) - I_r$ , whose support has basis- and basepoint-independent well-ordering. The trap is printed: a determinant-one rank-three family over  $\Gamma = \mathbb{Q}$  with one raw singular exponent and a *splitting* abelianized cocycle admits no splitting; one added term putting the forced descending set among the raw poles *removes* the obstruction; the true matrix logarithm is no repair, one coefficient off by a factor  $1/2$ . Also: rank-one recovery (cited, not reproved) of the scalar Cousin criterion of [surcomplex/global-divisors](#), the right torsor of all splittings, absoluteness under ordered value-group extension, and for every  $r \geq 3$  a cocycle trivial modulo the last central subgroup yet nonsplit in full positive rank. Part III (sections 19–27), from the third manuscript, glues by finite exponentials  $\exp(t^{\alpha_n} e^{1/(z-n)} N)$ ,  $N$  nilpotent,  $\alpha_1 > \alpha_2 > \dots > 0$ , putting descending exponents before ordinary *essential* singularities: determinant-one plane bundles, integral-trivial on bounded domains, whose Jordan-block members of rank  $r \geq 2$  stay indecomposable and not stably trivial over  $\mathcal{O}((t^\Gamma))$  and even over  $\text{Mer}((t^\Gamma))$ . It computes all sections and morphisms, classifies this one family (for fixed  $r \geq 2$ : profiles eventually equal integrally, eventually differing by one constant  $\delta \in \Gamma$  after localization), and proves a localization-stable cyclic dichotomy, continuum many classes in every noncyclic group and a tensor rule; the same manuscript supplies Theorem 28.4: for rational binary profiles neither isomorphism nor nonisomorphism is semidecidable. It does *not* settle Part I’s printed localization question: a merge observation (Corollary 27.2) makes Part I’s pole-data examples and the global-divisor report’s explicit plane Picard classes trivial over  $\text{Mer}((t^\Gamma))$ , deciding nothing over  $\mathcal{O}((t^\Gamma))$ . Shared lemmas are printed once; Part III’s labels are `nab:loc:`, the 169 earlier `nab:` labels intact. Not refereed, no Lean code, no priority certified. A companion placed in 21375f8, on inverting a monomial for the hidden rank-three cocycle, awaits integration.

### 36. Cofinality, factorization and scalar extension for entire Hahn functions at arbitrary rank

[surcomplex/entire-functions-at-arbitrary-rank/article.pdf](#)

from seven manuscripts, delivered 21–23 September 2026

Ordinary power series  $\sum a_n Z^n$  over a *fixed* Hahn field  $K_\Gamma = \mathbb{C}((t^\Gamma))$ ,  $\Gamma$  nonzero, set-sized, divisible and of arbitrary valuation rank; *entire* means strongly Hahn-summable at every point of that one field; for  $\Gamma = 0$  the ring is  $\mathbb{C}[Z]$ . The first two sources, which disclaim all of  $\mathbf{No}[i]$ , proved the same engine, printed once; both payloads are kept. **The trichotomy** (Theorem 1.5):  $E_\Gamma$  is always a GCD domain with units  $K_\Gamma^\times$ ; a polynomial PID when  $\text{cf}(\Gamma) > \aleph_0$ ; non-Noetherian Bézout with full Hermite interpolation when an *order unit* exists, sharply so (Theorem 4.1); **not** Bézout when  $\text{cf}(\Gamma) = \aleph_0$  without an order unit, two canonical products with disjoint simple zero sets and gcd 1 generating a proper ideal. **The exact jet image** (Theorem 1.6), from the third source, answering the question the first merge left open: jets on an infinite radially finite divisor  $D$  are realized iff on all late shells the shell’s Hermite polynomial has coefficients in the ring  $V_n$  that kills the Archimedean

scale of the shell radius; so  $E_\Gamma/(P_D)$  is an *algebraic* restricted product, not a topological or adelic one, with a sharp Bézout criterion. The fourth source, pinned before the third was merged, is recorded as an independent second proof of these theorems, its resolution claim stale; it adds finite-generator criteria, membership for one simple node per shell, the exact threshold for paired zeros, and, for one node per shell, all maximal ideals containing  $P_D$ : the evaluations and one per nonprincipal ultrafilter, a copy of  $\beta\mathbb{N}$ . None of this is a corona theorem; several nodes per shell and prime ideals stay open. **Scalar extension** (Theorem 1.7): for divisible  $\Gamma \subseteq \Delta$ , all nonpolynomial entire series share one exact strong evaluation domain in  $K_\Delta$ , the valuation ring coarsened by the convex hull  $H$  of  $\Gamma$ ; entireness survives exactly for cofinal extensions, no new zeros appear, and “not entire over  $K_\Delta$ ” is the coarse form of that exact domain. The fifth source (Section 11), without divisibility and over any trivially valued coefficient field, answers only the scalar-extension clause of the several-variable question: a whole-family entire series is strongly summable at a point with nonzero coordinates iff the weight image of its ordinary support is well ordered in  $\Delta/H$ ; under homogeneous-block summation, never called entire, the polynomial-direction loci of nonpolynomial entire series are, when  $\text{cf}(\Gamma) = \aleph_0$ , exactly the proper countable unions of projective algebraic sets. Extends **surcomplex/rank-one-berkovich**, which lands in the Bézout case; the whole-class rigidity re-derives, with credit, a conclusion already in **surcomplex/analysis** and **foundations-and-computation/foundations**; the restricted product here is not that of **surcomplex/global-divisors**. Rank is not the driver:  $\bigoplus_{\alpha < \omega_1} \mathbb{Q}e_\alpha$ , ordered by the largest nonzero index, admits only polynomials. No source was refereed or machine-checked, and priority is not certified. A sixth manuscript (Section 12) classifies every prime ideal above a canonical product with one simple node per shell: the quotient is zero-dimensional exactly when  $\Gamma$  has an order unit, otherwise continuum chains of primes appear, and a cofinal group extension can split one prime into a continuum-sized chain while the zeros are unchanged. A seventh manuscript (Sections 11.9–11.15) proves that, for  $d \geq 2$  variables and  $\text{cf}(\Gamma) = \aleph_0$ , every radially finite subset of  $K^d$  is carried into a coordinate line by an entire automorphism iff  $\Gamma$  has an order unit; without one, configurations of nonpositive support, the omnific integers of a workspace among them, still rectify, while explicit escaping infinitesimal simplices cannot be put in any hyperplane. On rectifiable configurations the fat-point ideals need exactly  $\binom{\bar{m}+d-1}{d-1}$  generators, and are not finitely generated when multiplicities are unbounded; this settles the ideal and value-interpolation clauses of the several-variable question for such configurations only. This addition lies outside the earlier proof-review scope.

### 37. Holonomic rigidity for entire Hahn functions: a sharp order-unit criterion for dilations

[surcomplex/holonomic-rigidity-for-entire-hahn-functions/article.pdf](#)

from thirteen manuscripts, delivered 22–23 September 2026

Power series over one fixed Hahn field  $K_\Gamma = k((t^\Gamma))$ , with  $\Gamma$  a nonzero set-sized ordered abelian group, not assumed divisible. Entire means strongly Hahn-summable at every point of that field; entireness can fail after enlarging the value group. The linear base is stated over  $k = \mathbb{C}$ , and the later algebraic-differential and mixed parts over any characteristic-zero field. The variable derivative  $D_z$  kills every scalar; it is not the Berarducci–Mantova derivation on surreal coefficients.

**Differential rigidity.** Entire D-finite series are polynomials, also in several variables. More precisely, a fixed nonzero linear operator has a uniform degree bound for formal solutions strongly evaluable at a sufficiently large argument. First-order algebraic differential equations also have only polynomial entire solutions. If  $\Gamma$  has no order unit, every nonpolynomial entire series is differentially transcendental in all orders. If  $\Gamma$  has an order unit, an explicit Chebyshev–theta series has minimal differential order three (Theorem H). The later chapters exclude order two for every value group (Theorem I), and order three with total jet degree at most three. Thus order three is the exact least possible order when an order unit exists, but a cubic jet equation cannot attain it. **Dilations and mixed equations.** For a parameter  $q$  of infinite multiplicative order, some nonzero polynomial-coefficient linear  $q$ -difference equation has a nonpolynomial entire solution exactly when  $|v(q)|$  is an order unit. A partial theta series witnesses existence; its precise

strong domain and the torsion boundary are also studied. For a finite list of multipliers with no root-of-unity quotient between distinct members, some mixed linear differential–dilation equation supported on that list has a nonpolynomial entire solution exactly when the absolute valuation of some quotient is an order unit. Two multipliers and first derivatives suffice for a witness. This answers the report’s mixed unit-valuation question; it does not decide solvability of a prescribed operator or classify torsion blocks. These are distinct from the Bézout and interpolation dichotomy in the entire-functions report. **Scope and verification.** The report also gives coefficient-field recovery, mixed-jet independence, uniform exclusion bounds and explicit infinite-rank witnesses. The source-08/09 linear arguments and source-10–12 nonlinear and coefficient-field parts have targeted manuscript reviews; the theta, low-order and newer mixed chapters await independent proof review. The Lean ledger maps support, evaluation, recurrence and valuation lemmas; it does not formalize the main classifications. Finite suites check identities and examples, not the infinite claims. No certified priority or independent refereeing is claimed. Polynomial composition, theta hierarchies, single scales and coefficient observables are now integrated. The latest recurrence chapter claims finite generation of the algebra containing unit-root coefficient sequences, and contrasts this with unrestricted single-coefficient observations; its 11 standard results and principal Theorem O await independent proof review.

### 38. Hahn–Tate uniformization at arbitrary valuation rank

[surcomplex/hahn-tate-uniformization/article.pdf](#)

from three manuscripts, delivered 22 September 2026

Tate elliptic curves over  $K = \mathbb{C}((t^\Gamma))$ , with  $\Gamma$  set-sized, of any rank and not necessarily divisible,  $q \in K$  of valuation  $\alpha > 0$ ,  $H_\alpha$  the convex subgroup of values bounded by an ordinary integer multiple of  $\alpha$ , and  $U_q = v^{-1}(H_\alpha)$ . **Theorem 1.2:** the bilateral theta family, and for  $u \in K^\times \setminus q^\mathbb{Z}$  each Tate coordinate family, is strongly Hahn-summable *exactly* on  $U_q$  — outside it the leading exponents contain a strictly decreasing sequence, so no cancellation repairs it — yet  $\Phi_q : U_q \rightarrow E_q(K)$  is a surjective homomorphism with kernel  $q^\mathbb{Z}$ . Source 01 is the base; source 03, written without knowledge of it, proved the theorem by another route, and both proofs are kept: Tate’s rank-one theorem in  $K_{H_\alpha}$  with coarse lifting (Sections 5–7), and an integral node chart with  $UV = q$  plus two further formal charts, with no rank-one completion (Section 8), valid over any coefficient field of characteristic zero (Theorem 8.13), hence over **No** for real  $q$  (Corollary 13.2). The domain half, the value sequence  $1 \rightarrow \mathcal{O}_v^\times \rightarrow E_q(K) \rightarrow H_\alpha/\mathbb{Z}\alpha \rightarrow 0$ , the good coarse reduction and the  $q = \omega^{-1}$  example are printed once. Also: exact support and valuation identities in the node chart; if  $H_\alpha \neq \Gamma$ , no homomorphism  $K^\times \rightarrow E_q(K)$  extending  $\Phi_q$  has kernel  $q^\mathbb{Z}$  (Theorem 10.5); recovery of  $q$  from  $j$  when  $v(j) < 0$  over any characteristic-zero coefficient field; and, over  $\mathbb{C}$ ,  $K(E_q[n]) = K_{\Gamma+\mathbb{Z}\alpha/n}$ . Every elliptic curve over **No**[ $i$ ] with  $v(j) < 0$  is isomorphic to an  $E_q$ . Part II, source 02, assumes monomial periods, coefficients in  $\mathbb{C}$  and  $\Gamma$  divisible where halves are used: every admissible lattice in a finite-dimensional higher-rank inner-product space has a finite positively generating set (Theorem 15.2), answering affirmatively Question 11.14 of Amini–Nicolussi *as printed* in 2025; an exact flag criterion for quadratic lattice families, support order type  $\omega^s$ , the exact multiscale theta domain, finite minimum certificates, and lifting of smooth initial zeros. **The bounded-period quotient is not new:** Molcho–Wise’s bounded monodromy and logarithmic Tate curve are a precedent, correcting source 01’s novelty framing, and no functorial comparison is proved. No abelian variety is uniformized,  $v(j) \geq 0$  is not treated; not refereed, no Lean verification, priority not certified by the targeted searches. Neighbours: [surcomplex/rank-one-berkovich](#) (Tate *algebras*, not Tate curves) and, for Part II, [surcomplex/wick-summability-certificates](#).

### 39. Infinite-dimensional Hahn spectral theory: infinitesimal thickening, range defects, and algebra-relative spectra

[surcomplex/infinite-dimensional-hahn-spectral-theory/article.pdf](#)

from four manuscripts, delivered 22 September 2026

Two manuscripts over one Hahn field  $K = \mathbb{C}((t^\Gamma))$ , printed as *two parts*: *spectrum* means different things on their two vector spaces and in their different operator algebras. Section 3 sets the two side by side on  $\text{diag}(1/n)$ ; each part is self-contained given Section 2. **Part I**:  $H$  an ordinary complex Hilbert space,  $\Gamma$  nonzero, set-sized and ordered abelian, without divisibility (only Section 12.1 assumes  $\Gamma \subseteq \mathbb{R}$ ), and  $H((t^\Gamma))$  with the coefficientwise-convolution inner product. Every everywhere-defined adjointable operator, with no continuity assumed, is a Hahn series of ordinary bounded operators with one common well-ordered support (Theorem 6.1). For ordinary bounded normal  $T$  the failure-of-bijectivity spectrum is  $\sigma_{\mathbb{C}}(T) \cup (\sigma_{\mathbb{C}}(T)' + \mathfrak{m}_K)$ , with no new eigenvalues (Theorems 8.2, 8.3): accumulation points thicken into infinitesimal monads, isolated points do not. An injective linear  $S : V \rightarrow V$  keeps  $\text{coker}(S+E) \cong W((t^\Gamma))$ , where  $V = \text{Ran } S \oplus W$ , under every  $E$  of strictly positive order (Theorem 7.1). With  $De_n = e_n/n$  on  $\ell^2$  and  $\eta > 0$ ,  $C = D^2 + t^{2\eta}I$  is self-adjoint and coercive with the infinitesimal constant  $t^{2\eta}$ , yet its cokernel has dimension at least  $2^{\aleph_0}$  (Theorem 11.3), and at every rank its range is closed and proper with zero orthogonal complement (Theorem 12.2). A least field-valued norm bound exists exactly when  $T$  attains its ordinary norm (Theorem 13.2). **Part II** also allows arbitrary ordered value groups. In  $\mathcal{A}_I = \text{RCF}_I(\mathbb{C})((t^\Gamma))$  acting on  $\mathbb{C}^{(I)}((t^\Gamma))$ ,  $A = \text{diag}(d_i) + B$ , with pairwise distinct ordinary real  $d_i$  and  $B = B^*$  of globally well-ordered positive support, has a unique near-identity normalized Hahn-unitary diagonalization with corrections in the monoid generated by  $\text{supp } B$  (Theorem 18.2, Corollary 18.3), exact synthesis, classified commutant and spectral projections, and  $\sigma_{\mathcal{A}_I}(A) = \{\lambda_i\}$  even when the  $d_i$  accumulate (Theorem 20.2) — a spectrum relative to that algebra, not a Banach-algebra spectrum. Closed weighted walks control every eigenvalue coefficient, and the one-scale finite-section bound  $(2r+2)\eta$  (Corollary 21.5) cannot be improved in general (Theorem 21.6). *Merged*: the Hahn–Neumann support lemma is stated once, and Part II’s proof of Higman’s lemma is not reprinted, the identical argument being `spec:lem:words` and `spec:lem:neumann` of `surcomplex/spectral-theory`. The proof review removes the unused divisibility assumption of Part I and proves range closedness by its continuous defect projection at every rank; neither part uses a theorem of the other. It supplies the first two items of that report’s closing non-claim and the compact half of the third, compact meaning compact on  $H$ . Part II supplies a vectorwise Hahn-additive projection calculus, with a rank-one counterexample to pointwise valuation convergence. No trace-class theory or general topological spectral-measure theorem is provided; `surcomplex/hahn-herglotz-positivity` gives a separate obstruction to a positive coefficientwise circle measure. Both suites report 218 checks, by coincidence; the assertions differ. Priority not certified; not refereed; no proof-assistant check. Two later manuscripts add Parts III and IV. Part III computes the spectrum of a constant element relative to  $\mathfrak{A}((t^\Gamma))$  for any unital algebra  $\mathfrak{A}$  through Drazin invertibility (Theorem 32.1): it is Part I’s adjointable spectrum for  $B(H)$  and Part II’s for row- and column-finite matrices, and recovers only that half of Part I’s theorem. Part IV, with  $\Gamma$  divisible, treats compact self-adjoint operators under noncommuting positive-valuation perturbations: finite Hermitian clusters at nonzero eigenvalues, the whole monad of zero in the spectrum, and no global bounded-coefficient spectral theorem; its coefficientwise Fredholm determinant is not a trace-class theory.

### 40. Expanding polynomial dynamics over surreal and surcomplex fields: exact symbolic fibers and the order-unit obstruction

[surcomplex/expanding-polynomial-dynamics/article.pdf](#)

from one manuscript, delivered 22 September 2026

Iteration of  $F = q^{-1}P$  over  $K = k((t^\Gamma))$ ,  $k = \mathbb{R}$  or  $\mathbb{C}$ , with  $\Gamma \neq 0$  any set-sized ordered abelian group — no divisibility, no algebraic closedness — where  $\kappa = v(q) > 0$ ,  $P$  is integral of degree  $d \geq 2$ , and its reduction keeps degree  $d$  and has  $d$  distinct roots in  $k$ . With  $I_\kappa = \{h : v(h) > n\kappa \text{ for all } n \in \mathbb{N}\}$ : every word  $s \in \{1, \dots, d\}^{\mathbb{N}}$  has a canonical center  $x_s$  with  $F(x_s) = x_{\sigma s}$ , built by formal algebra and

strong Hahn evaluation under one support certificate for all words; the integral nonescaping set is  $\bigsqcup_s(x_s + I_\kappa)$ ; and words first differing at index  $m$  give points at valuation distance exactly  $m\kappa$ . So a word determines one point *iff*  $\kappa$  is an order unit — a property of the scale, not of the rank. The preperiodic points are exactly the centers of eventually periodic words, and each  $F^{m+p} - F^m$  has  $d^{m+p}$  simple roots in  $K$ . **Order-unit dichotomy:** for an order unit,  $s \mapsto x_s$  is a homeomorphism from the full shift onto a compact perfect set; otherwise the fibers are open and the iterates uniformly equicontinuous, though the Hausdorff quotient of the  $\kappa$ -scale uniformity is still the full shift, and a separate compact-subsystem rigidity theorem makes every compact forward-invariant subset finite. Also scalar extension (fibers can thicken, no new preperiodic points), deformations with exact leading response, the exact valuation-bounded locus, a no-order-unit equicontinuity criterion for all polynomials, and  $F(x) = \omega(x^2 - 1)$  in  $\mathbf{No}[i]$  with real centers. One source, not a merge: nothing was selected, and at placement only terminology changed, and a later proof review clarified degree preservation, formal substitution and topology. The manuscript’s “cofinal scale” is an order unit here, since **surcomplex/entire-functions-at-arbitrary-rank** reserves cofinality for  $\text{cf}(\Gamma)$ . Valuation-expanding is not repelling and no Julia/Fatou theory is built, so neither exclusion of **surcomplex/dynamics-and-normal-forms** is answered. In  $\mathbf{No}[i]$  the set-level discreteness and compactness clauses are special cases of the closed-and-discrete theorem of **surcomplex/analysis**; the substance lies in set-sized workspaces. The order-unit case has the classical symbolic Cantor model even at higher rank; every positive scale is an order unit in the field of **surcomplex/rank-one-berkovich**;  $H_\kappa$  is the  $H_\alpha$  of **surcomplex/hahn-tate-uniformization**, a shared subgroup, not a shared theorem. Repeated residue roots and rational maps are not covered. Not refereed; priority not certified. Lean proves the scale subgroup and error-ideal lemma, including the valuation-ball conditions for clopenness and the coarsened valuation ring. The dynamical theorems remain pending; the 38,139 exact finite checks prove no infinite statement.

#### 41. Positivity without a positive measure: Hahn–Herglotz normalization and strict moment hierarchies in surcomplex analysis

[surcomplex/hahn-herglotz-positivity/article.pdf](#)

from one manuscript, delivered 22 September 2026

Over  $\mathbb{R}((t^\Gamma))$  and  $\mathbb{C}((t^\Gamma))$ , with  $\Gamma$  a nonzero set-sized ordered abelian group, and by transport over set-sized subfields of  $\mathbf{No}[i]$ , finite Toeplitz positivity, positive real part on the halo of an ordinary domain, and positive-measure representation on the *ordinary* circle come apart. Measures are real-valued and *coefficientwise*: an ordinary finite signed regular Borel measure at each exponent of one well-ordered support, additive exponent by exponent, without requiring strong summability; moments are Fourier moments. For divisible  $\Gamma$ , assumed only for the congruence in Theorem 3.3, a matrix coherent function with  $\text{Re } H \succeq 0$  at ordinary points is, after subtracting a constant  $iB$  with  $B$  Hermitian, constant-congruent to  $G \oplus 0$  with  $G(a_0) = I$ ,  $\text{Re } G_0 \succ 0$  and otherwise unrestricted higher coefficients. For every nonzero  $\Gamma$ , ordinary and halo positivity agree, kernels are constant, and Harnack holds with every ordinary  $C > C_0$  but need not with the sharp classical constant (Theorem 3.5). Theorem 4.2:  $\sum \nu_\gamma t^\gamma$  is positive iff each  $\nu_{\bar{\gamma}}$  vanishes on the common null ideal of the earlier  $|\nu_\beta|$ . Corollary 5.2 gives a strictly positive, unital, pointwise completely positive functional with no positive representing measure. The moments  $c_0 = 1$ ,  $c_n = -\varepsilon$  have  $\det T_N = (1 + \varepsilon)^N(1 - N\varepsilon) > 0$  for every ordinary  $N$ , yet their unique signed measure is  $(1 + \varepsilon)m - \varepsilon\delta_1$ , possible only because  $\varepsilon$  is infinitesimal; positive quadratures converge coefficientwise weak-\* to it, the Schur parameters are  $-\varepsilon/(1 - n\varepsilon)$ , and the Schur iterates have internal poles of standard part 1. Theorem 7.1 separates the positive-measure, signed-measure, distribution, Toeplitz-only and halo-Herglotz classes strictly, with exponents  $0, \eta$  only. A single source; the main equivalence of Theorem 4.2 is proved once, in **surreal/hahn-valued-measures-and-probability** as `meas:thm:nullideal`, found independently by a source of that report in the same batch; this report proves the translation between the two forms and the control-measure reformulation, and keeps its source’s uncountable example showing that the latter cannot replace the null ideal. In that report’s strong class  $\omega + 1$  is the least support order type letting nonnegative finite tests hide a negative event; in the coefficientwise class here

two exponents suffice. The halo is that of `surcomplex/analysis`, whose `c:p5:positivity` is the mechanism of Theorem 5.1(b); Corollary 6.5 adds only negative information to `surcomplex/spectral-theory` and `surcomplex/infinite-dimensional-hahn-spectral-theory`. Among 26 non-claims: nothing on enlarged circles, finitely additive or nonstandard representations, or a matrix null-ideal criterion. A candidate contribution; priority not certified and not refereed. Lean proves the null-ideal and two-scale criteria and finite algebraic clauses of the negative-atom, quadrature and Schur examples; the remaining analytic and representation claims are pending.

#### 42. Sharp moment reconstruction at surreal scales: collision geometry, optimal precision, and Hahn-supported inversion

[surcomplex/prony-reconstruction-at-surreal-scales/article.pdf](#)

from one manuscript, delivered 22 September 2026

Recovery of  $n$  distinct integral nodes  $a_i$  and nonzero weights  $w_i$ , of any valuation, from the power moments  $m_k = \sum_i w_i a_i^k$ ,  $0 \leq k < 2n$ , over  $\mathbb{R}((t^\Gamma))$  or  $\mathbb{C}((t^\Gamma))$  with  $\Gamma$  any nontrivial ordered abelian group, not assumed divisible. No measure or positivity is involved; since neighbouring reports use “moment” otherwise, the directory is named after Prony. With  $d_i = \sum_{j \neq i} v(a_i - a_j)$ ,  $\delta_i = \max_{j \neq i} v(a_i - a_j)$ ,  $E_i = v(w_i) + 2d_i$  and  $\Theta = \max_i (E_i + \delta_i)$ , Theorem 1.1: if all  $2n$  moment errors have valuation at least  $\kappa > \Theta$ , there is a unique regular  $n$ -node realization, uniquely labelled by the strict nearest-neighbour balls  $v(\hat{a}_i - a_i) > \delta_i$ , with node errors of valuation  $\geq \kappa - E_i$  and weight errors  $\geq \kappa - 2d_i - \delta_i > v(w_i)$ ; perturbing only  $m_{2n-1}$  attains every node bound at once, and at valuation exactly  $\Theta$  leaves no regular realization with that labelling, for *every* configuration. Optimality is for this labelling, not for bare existence; failure at  $\Theta$  need not be a collision or a loss of positivity. Also: exact Gauss valuations  $-2d_i$  and  $-2d_i - q_i$  of the inverse Hermite rows, with a cancellation depth  $q_i$  that the separation tree does not determine; the  $q_i$ -improved weight bound, proved for the tangent map, and for the whole nonlinear set only under the extra finite criterion (not claimed optimal) of Proposition 7.2, an arbitrary-rank form of the Caruso–Roe–Vaccon precision-lattice principle, with Proposition 9.1 showing that the improvement can fail nonlinearly; a *sufficient* directed-path certificate for nonuniform precisions, valid also for nondivisible  $\Gamma$  by auxiliary scaling in its divisible hull and descent to the original field; and transfer to finite configurations in **No** and **No**[ $i$ ]. One manuscript, not a merge; the only choices were editorial (title kept, labels prefixed `prony:`, notation aligned with `surcomplex/polynomial-algebra`). Remark 6.2, added on placement and proving no new theorem, compares  $\Theta = \max_i ((v(w_i) + d_i) + (d_i + \delta_i))$  with that report’s coefficient threshold  $T(P) = \max_i (d_i + \delta_i)$ . Only the classical Hankel identity is shared with `surcomplex/spectral-theory`; the Prony-type lemma of `surreal/hahn-valued-measures-and-probability` is cross-referenced, not merged. Classical Prony, Hermite and precision-tracking material is credited, as are the  $w_i P'(a_i)^2$  denominator and the last-moment direction. Not an unconditional algorithm on surreal inputs; confluent and infinite configurations are out of scope; the 10,052 exact SymPy assertions check finite instances only. Priority is not certified; not refereed. The Lean ledger records classical uniqueness, cofactor orthogonality, the exact last-moment annihilator and algebraic local-root identities; valuation localization and the remaining precision estimates are pending.

#### 43. Finite certificates for surcomplex Wick summability: valuation balancing, connected obstructions, and multiscale Gaussian perturbations

[surcomplex/wick-summability-certificates/article.pdf](#)

from one manuscript, delivered 22 September 2026

Formal Gaussian perturbation theory over  $K = \mathbb{C}((t^\Gamma))$ ,  $\Gamma \neq 0$  set-sized, *divisible* and of arbitrary rank: a finite interaction  $P = \sum_a g_a x^{\alpha_a}$ ,  $g_a \in K^\times$ ,  $\alpha_a \in \mathbb{N}^d \setminus \{0\}$ , and a symmetric *bilinear* covariance  $C$ , possibly singular and not necessarily real or positive; the expectation is a finite sum over pairings. With  $\lambda_a = v(g_a)$  and  $c_{ij} = v(C_{ij})$  on nonzero entries, Theorem 6.1: the vacuum Wick diagram family is strongly summable iff its weight is positive on the finite Hilbert basis of  $S = \{(m, k) : Am = Dk\}$ , iff some  $p \in \Gamma^d$  has  $\lambda_a + \alpha_a \cdot p > 0$  and  $c_{ij} - p_i - p_j > 0$  (a diagonal monomial rescaling making all couplings and nonzero covariances infinitesimal), iff

every observable sector, equivalently any one pairable sector, is summable; Hahn-unit tails are arbitrary. The bridge is a strict alternative over any divisible ordered group, by Fourier–Motzkin elimination without real separation (Theorem 5.1); failure yields a nonnegative integer certificate in  $S \setminus \{0\}$  of nonpositive weight, whose multiples obstruct. With every valence at least 2 the connected vacuum diagrams have the same domain (Theorem 7.3; valence 1 fails, Example 7.5). Positive-definite real covariance, or diagonal covariance with nonzero diagonal, reduces the test to  $\lambda_a + \frac{1}{2} \sum_i \alpha_i^a c_{ii} > 0$  per monomial (Theorem 8.2), giving a sharp multiscale stationary-phase threshold (Corollary 8.3); in a mixed quartic each interaction alone is admissible but the pair is not. **Diagramwise only:** for  $C = I_2$ ,  $P = t^{-1}(x + iy)^4$  the order-grouped sum is 1 while the diagram family is not summable, and the criterion is not invariant under general complex linear coordinate changes. A Hahn value is not an integral, summability is not convergence of truncations in rank two (Example 9.4), and no physics is claimed. Exact solver over lexicographic  $\mathbb{Q}^r$ ; 7,684 finite checks at ranks 1–3. Single source, not a merge; placement added Sections 1.3, 13.4 and 13.5, and Section 13.1 corrects the source’s report that a repository search for *Gaussian* found nothing. Where **physics/surreal-scalars-and-spacetime** cautions that admissibility is not automatic, this report decides it exactly for one class, and answers none of that report’s questions; the closest relative is the multiscale theta series of **surcomplex/hahn-tate-uniformization**, also decided by a finite certificate but by different mathematics; neither uses the other. Classical inputs credited; priority provisional; not refereed. The Lean ledger covers the positive-support, finite-semigroup and valuation Cauchy–Schwarz lemmas; the main summability and connected-domain theorems remain pending.

#### 44. Three duals at surreal scales: completion, strong maps and valuation continuity [surcomplex/three-duals-of-hahn-vector-spaces/article.pdf](#)

from one manuscript, delivered 22 September 2026

For a  $k$ -vector space  $V$  and  $V_\Gamma = V((t^\Gamma))$  over  $k((t^\Gamma))$ , three objects are kept apart: the algebraic scalar extension, its valuation completion, and its closure under strong Hahn summation. The completion consists exactly of the series whose coefficients below each valuation cut span a finite-dimensional space; in infinite coefficient dimension it is the whole Hahn space exactly when  $\Gamma \cong \mathbb{Z}$ , and it is strictly larger than the algebraic extension exactly when  $\Gamma$  has countable cofinality; a cofinal exponent enlargement preserves this and a noncofinal one collapses it. Every strong linear operator is one Hahn series of arbitrary  $k$ -linear coefficient maps. For infinite-dimensional Hilbert  $H$  and noncyclic  $\Gamma$  there are three strictly nested duals — inner-product-represented, strong, valuation-continuous — the first gap being the Hahn extension of  $H^\# / H'$ ; all strong maps are bounded by a positive real Hahn scalar; and a choice-based hyperplane has a strong bounded projection but zero orthogonal complement. It works on the Hahn–Hilbert space of **surcomplex/infinite-dimensional-hahn-spectral-theory**, whose non-represented functional it shows to be strong, and whose adjointable operators sit inside its strong ones. One manuscript; some cited literature was not available to its author and is attributed, not verified; not refereed, no Lean verification.

#### 45. Hidden negative directions of Hermitian forms

[surcomplex/hidden-negative-hermitian-directions/article.pdf](#)

from one manuscript, delivered 22 September 2026

For divisible  $\Gamma \neq 0$  let  $\mathcal{P}_\Gamma \subset \mathbb{C}((t^\Gamma))$  be the series supported in finitely generated subgroups: an algebraically closed subfield containing every monomial. For every  $r \geq 1$  an explicit Hermitian matrix of size  $r + 1$  is strictly positive at every nonzero vector over  $\mathcal{P}_\Gamma$ , and even over its valuation closure, yet has inertia  $(1, r, 0)$  over  $\mathbb{C}((t^\Gamma))$ ; so the exponent group in the Hermitian spectral theorem of **surcomplex/spectral-theory** cannot in general be replaced by a finitely generated one. “Positive semidefinite” always means nonnegative at *every* vector over the full field. The engine is prime-denominator tails  $\sum_p t^{\delta(1-1/p)}$ : every finite-lattice approximant misses below valuation  $\delta$ , matching  $N$  terms algebraically needs degree exactly  $\prod_{j \leq N} p_j$ , and almost-disjoint prime sets give continuum many independent tails. A second theorem characterizes positivity of a two-scale matrix-valued measure  $M_0 + \varepsilon M_1$  by scalar null ideals and a vector-valued absolute-continuity condition

— the two-exponent case of the matrix criterion that `surcomplex/hahn-herglotz-positivity` excludes; more exponents stay open. Transported to surreal normal forms. One manuscript; background inputs are named as such; not refereed, no Lean verification.

#### 46. Automorphisms of the `surcomplex` field

`surcomplex/surcomplex-field-automorphisms/article.pdf`

from three manuscripts, integrated through fd91f9b

“Automorphism of  $\mathbf{No}[i] = \mathbf{No}(i)$ ” has several inequivalent meanings, and this report keeps six apart: plain field, valued, value-fixing, 1-automorphisms, exponential, and those preserving the logarithmic modulus. The automorphisms fixing  $\mathbf{No}$  pointwise are the identity and conjugation; those preserving  $\mathbf{No}$  setwise form  $\text{Aut}(\mathbf{No}) \times C_2$ , a proper subgroup of all field automorphisms, as Hahn-series phase twists show by fixing every ordinary complex number and every value while moving the surreal real axis. Also the coefficient, character, value-group and leading-term-kernel layers of the valued automorphism group, automorphisms moving ordinary real constants by infinitesimals, zero-derivative field automorphisms and real forms. The exponential results — the amplification lemma, faithfulness, the negative answer to Kaplan–Krapp–Serra Question 5.4 and the logarithmic-modulus classification — are reprinted with attribution from `surreal/exponential-automorphism-rigidity`; here  $\text{Aut}_L(\mathbf{No}[i])$  is observed to be a proper subgroup of the real-axis stabilizer. Readings of the Kaplan–Krapp–Serra paper beyond Question 5.4 are the source’s and were not rechecked. Not refereed; generic Lean coverage uses the rigidity report’s labels. The later finite-symmetry and Gaussian-omnific real-form parts expand the main text. Source 03 adds fourteen standard results and remains outside the earlier Sections 1–15 proof review.

#### 47. The first- $\kappa$ coefficients: omitted types and completion in Hahn fields

`surcomplex/first-kappa-coefficients/article.pdf`

from `surreal_first_kappa_types.zip`, delivered 22 September 2026

Inside a full Hahn field  $k((t^\Gamma))$ ,  $k = \mathbb{R}$  or  $\mathbb{C}$ , take the series with fewer than  $\kappa$  nonzero terms, for an uncountable and possibly singular cardinal  $\kappa$ . Which one-variable types does the full field realize that this subfield omits? Exactly those coded by the first  $\kappa$  nonzero terms of a series: these terms determine its complete ordered-field type in the real case and its valued-field type in the complex case, and the least number of formulas witnessing omission is  $\text{cf}(\kappa)$ , while in the pure complex field language all missing elements share one type. The same invariant gives the approximation cut, the least empty nest of closed balls, the valuation completion (the subfield is complete exactly when  $\text{cf}(\Gamma) \neq \text{cf}(\kappa)$ , and never spherically complete) and the valuation loss of linear retractions; Conway normal forms give surreal examples at  $\kappa = \aleph_\omega$ . The countable-support analogues in `surcomplex/three-duals-of-hahn-vector-spaces` and `surcomplex/rank-one-berkovich` are cross-referenced. The constructions and quantifier-elimination inputs are established; the proposed contribution is the joint classification. Not refereed, no Lean verification, limited priority audit.

#### 48. A single dilation recovers Hahn support

`surcomplex/single-dilation-hahn-support/article.pdf`

from `surreal_dilation_support.zip`, delivered 22 September 2026

Can one field automorphism make the Hahn structure of  $k((t^\Gamma))$  visible? For every coefficient-fixing monomial automorphism  $\sigma(t^g) = \chi(g)t^{\theta g}$  the report constructs a strong partial inverse of  $\sigma - \lambda$ , obstructed only at exponents fixed by  $\theta$  where  $\chi = \lambda$ , and for finitely many commuting ones a deformation retraction of the Koszul complex onto its joint-resonant coefficients, computing additive and principal-unit multiplicative cohomology. The main application: the single dilation  $S_2(\sum a_g t^g) = \sum a_g t^{2g}$  defines, in the language of fields with one automorphism, the coefficient field, every coefficient, the canonical monomials, the valuation ring and truncation. Hence its centralizer among *all* field automorphisms is exactly the untwisted coefficient–exponent lifts, full monadic second-order arithmetic is interpreted (so the theory is undecidable, with a  $\text{TP}_2$  witness), and all nonidentity rational dilations give the same definable data though no two are conjugate.

The dilation is the  $S_a$  of [surcomplex/surcomplex-field-automorphisms](#), whose questions it does not answer. Elementwise results reach **No** and **No**[ $i$ ] through Conway normal forms, with proper-class qualifications stated. Not refereed, no Lean verification.

#### 49. Gamma and zeta over the surcomplex numbers

[surcomplex/gamma-and-zeta-functions/article.pdf](#)

from six manuscripts, delivered 22 September 2026

What can  $\Gamma(z)$  and  $\zeta(z)$  mean when  $z \in \mathbf{No}[i]$ ? Not one function: Taylor–Laurent evaluation at finite points, strong Hahn summation at infinity and the choice of a global exponential are kept apart throughout. At finite points the lifts have exactly the classical zeros and poles, so their Riemann hypothesis is the classical one, and stability of every finite zero under symmetric infinitesimal deformations is equivalent to RH with all zeros simple. At positive infinite real part every formal Dirichlet series is strongly summable, evaluation is an injective algebra map with exact Euler products and no zeros of zeta, every fibre  $\zeta = 1 + u$  is described explicitly, and zeta jets at dilations and at real shifts are algebraically independent over large fields (the prime lemma being Pong’s). Stirling, a phase-free ratio cocycle and Hurwitz zeta at an infinite parameter follow. With any global phase, Gamma, zeta and  $\xi$  exist on the whole horizontal tube; reflection then forces Gamma poles and zeta zeros on the phase’s infinite periods, and with the Ehrlich–Kaplan exponential zeta vanishes at the negative even omnific integers, while  $\xi$  keeps exactly the ordinary zeros. At infinite imaginary part no exponential extending the finite-phase one gives reflection anywhere on the vertical strip, and resonant phases obstruct a coherent continuation; elementary extensions instead transfer classical zero-free regions, density bounds and de Bruijn–Newman results. *Merged from six manuscripts on the same questions; each was audited alone and all were compared claim by claim. No two contradict each other once domain, exponential, summation notion and deformation class are aligned; the reconciliation is printed at each point of use, and fifty-three minor source errors are fixed and listed.* Results the merge itself proves are marked. No proof of classical RH, no canonical Gamma or zeta on all of **No**[ $i$ ], not refereed, no Lean verification.

#### 50. Hilbert geometry at surreal scales

[surcomplex/hahn-hilbert-geometry/article.pdf](#)

from three manuscripts, integrated through 5ca4355

Which closed subspaces of a Hahn–Hilbert space  $H((t^\Gamma))$  have orthogonal complements? Exactly the infinitesimal unitary rotations of ordinary closed subspaces, equivalently the graphs of positive-order operators; a closed residue is necessary but not sufficient. Every unitary factors uniquely as an ordinary unitary times  $\exp X$  with  $X$  skew of positive order. The graph of an amplified operator  $aT$  splits iff  $a = 0$ ,  $v(a) \geq 0$  or  $T$  has closed range, and explicit graphs have no nearest point, even after enlarging  $\Gamma$ . The split projections form an orthomodular poset that is a lattice iff  $\dim H < \infty$ ; countable orthogonal families of rank-one projections have no join, and the operator algebra is not Rickart. A positive metric is isometric to the canonical one iff its ordinary part is bounded below by a positive constant. For  $A = T_0 + E$ ,  $T_0$  Fredholm and  $E$  of positive valuation, kernel and range split,  $A^\dagger$  exists with an explicit valuation, and every least-squares problem has a minimum-norm solution. Through scale extension the geometry, counterexamples and metric rigidity hold over **No** and **No**[ $i$ ] for class maps with class adjoints. *The first two manuscripts share results; the base has the weaker hypotheses on every shared theorem.* The adjoint theory is that of [surcomplex/infinite-dimensional-hahn-spectral-theory](#). Not claimed: a spectral theorem, a splitting criterion from closed residue alone, a physical interpretation. Proof review pending; not refereed, no Lean formalization, priority not certified. The third manuscript adds an exact criterion for infinite simultaneous unitary straightening; its fifteen new standard results await independent review and Lean formalization.

### 51. Dilation rigidity of surreal numbers and omnific integers

[surcomplex/autonomous-dilation-relations/article.pdf](#)

from three manuscripts, assembled through 1ef6bba, 23 September 2026

For  $d \geq 2$  the exponent dilation  $S_d$  fixes coefficients and multiplies normal-form exponents by  $d$ ; it is not  $y \mapsto y^d$ . The placed manuscript claims that a nonconstant Hahn series satisfying a nonzero constant-coefficient relation  $F(y, S_dy) = 0$  lies in one cyclic Laurent subfield, with a ramification bound  $\deg_Y F$ , and deduces that infinite-support omnific integers are algebraically independent from each of their nonidentity integer dilates. Two later manuscripts add 33 standard results: sufficient support rank forces independence of distinct positive dilates, giving exact autonomous order for finite support, with relative versions and infinite-support constructions. Independent proof review and formalization remain pending. Not refereed, no Lean formalization, priority not certified.

## 3 Surquaternions

One report, which is where commutativity stops. It is filed on its own because commutative arguments from `docs/surcomplex` require review on moving to a division algebra: conjugation, multiplicativity of the modulus, valuation and leading coefficients, polynomial evaluation, matrix adjoints, exponential laws and every chain rule need their scalar and slice hypotheses checked. The report proves the additional noncommutative assertions with factor order retained.

### 52. Surquaternions: algebra, infinitesimal geometry, and support-controlled analysis

[surquaternions/surquaternions/article.pdf](#)

from `surquaternions_article.zip` + `surquaternions_article(1).zip`

Hamilton’s quaternion division algebra over the surreals,  $\mathbb{H}_{\mathbf{No}} = \mathbf{No} + \mathbf{No}i + \mathbf{No}j + \mathbf{No}k$ , with central surreal coefficients, in three levels that are never silently mixed. *Finite algebra over an arbitrary real closed field*: division, conjugation, the field-valued modulus, complex slices and centralizers, conjugacy spheres, explicit square roots, the spin/rotation description with two independent surjectivity proofs, Cayley and stereographic presentations of one rational chart, quaternionic Möbius maps, one-sided polynomials with a constructive fundamental theorem and a complete zero-class algorithm, and finite-dimensional Hermitian spectral theory with SVD and polar decomposition. *Set-sized Hahn workspaces*  $D_\Gamma = \mathbb{H}(\mathbb{R}((t^\Gamma)))$ : normal forms with a **noncommutative** residue algebra, exact multiplicative valuation and leading coefficients, the all-lengths Neumann support lemma proved through Higman’s finite-word lemma, the exact geometric inverse with its truncation certificate, the infinitesimal Hahn exponential and logarithm, BCH at positive valuation, the filtered nonabelian rotation group with its graded bracket, a spectral-projector estimate at arbitrary surreal gap scale with a sharp two-scale example, support-controlled implicit lifting, and an exact square-root conditioning analysis. *The full surreal class*: finite angles at arbitrary radius, the Ehrlich–Kaplan global phase normalization, the global radial exponential with its complete logarithm fibres and four-coordinate differential — rank-two critical spheres at positive radii in  $\pi\mathbf{Oz}$ ,  $\mathbf{Oz}$  the omnific integers, the infinite radius  $\omega$  among them — the componentwise Berarducci–Mantova derivation with a no-oscillation theorem and its chain-rule corollary, and coefficientwise slice, Cauchy and Fueter analysis on explicitly named coefficient spaces. **Section 2 is the ledger and should be read first**: three exponentials and four derivative-like operators carry overlapping names here, each satisfying a *different* law on a *different* domain, and the section fixes one name and one symbol per object and tabulates which satisfies which. It also explains why this report’s statement that the radial exponential is *not* an additive-to-multiplicative homomorphism is consistent with the spacetime physics report’s correct claim that the *commutative* Ehrlich–Kaplan exponential *is* one: the restriction to any single slice is exactly that commutative exponential and the law holds there, and it can fail across *noncommuting* arguments, where no common slice exists — witness  $\text{Exp}(\omega i + \pi j) = 1 + (\pi^2/2\omega)i + O(\omega^{-2})$  against  $\text{Exp}(\omega i)\text{Exp}(\pi j) = -1$ . A merged sentence blurring the two would manufacture a false

contradiction across the collection. *Merged; roughly two thirds of the mathematics was common to both sources and is printed once.* Every `\label` in its source carries the prefix `squat:`. It makes no claim about any empirical subject.

## 4 Physics

Two reports, the only ones whose subject is not mathematics alone. Both separate mathematical models from physical evidence, and both tag every claim as an exact identity, a conditional theorem, an assessment or an import. The relativity report asks what surreal scalars do and do not buy near a gravitational singularity; the quantum and gauge report, merged from four manuscripts, asks the same of finite quantum operations and gauge models. Neither claims an experimental prediction.

### 53. Surreal scalars in theoretical physics: black-hole singularities, asymptotic structure, and the limits of replacing the scalar field

[physics/surreal-scalars-and-spacetime/article.pdf](#)

from three manuscripts, delivered 21 September 2026

**The sentence this report exists to refuse, in any form, is that surreal numbers resolve the black-hole singularity.** Surreal and surcomplex scalars make the hierarchy of quantities near a gravitational singularity exactly representable; they do *not* make the spacetime regular. Four facts, each re-derived from the metric for the merge rather than copied from a source: the Kretschmann scalar of Schwarzschild is exactly  $K = 48M^2/r^6$ ; under a formal radial change  $r = s^q w(s)$ , with integer  $q \geq 1$ ,  $w(0) \neq 0$  and  $M \neq 0$ , and  $w \in \mathbb{R}[[s]]$ , the pole order becomes exactly  $6q$  with leading coefficient  $48M^2 w(0)^{-6}$ . The recorded checks use  $w(s) = 1 + s$ , whose constant term is one; this normalized case preserves  $48M^2$ . No substitution of the stated form removes the pole; the proper time from rest at  $R$  to  $r = 0$  is  $(\pi/2)\sqrt{R^3/2M}$ , which for *infinitesimal*  $R$  is infinitesimal and hence still finite, so the infall terminates either way; and the tidal component  $2M(2M - r)/r^4$  is unlimited at infinitesimal  $r$ . The exact identity  $r^6 K = 48m^2$  has *no* field-valued solution at  $r = 0$  in any characteristic-zero field, not merely in  $\mathbb{R}$ : evaluating at a nonzero infinitesimal radius names another point of a punctured domain and supplies no missing endpoint, no continuation law and no observable rule. What is true, and is what the report is for, is that the surreal/Hahn language records the power *and* the coefficient of every divergence exactly, keeps competing scales separate, and detects when an expansion has been used outside its regime — a gain in representation and nothing more. **The claim-status firewall** is the rest of its value: every theorem header and every substantive paragraph carries one of four visible tags — exact symbolic identity, checked exactly and never to a tolerance, which is the entire content of what the programs establish; conditional theorem, with hypotheses stated in full and *established for no physical model*, nothing in the tier machine-checked; assessment, meaning an argument and not a result; and imported, a provenance marker rather than a fourth tier. The article’s own tag counts are 53, 51, 140 and 34. *Merged; duplication between the three sources ran to about two thirds, all three independently rebuilding the static-spherical connection and the full Riemann tensor and re-deriving the same Schwarzschild quantities from the metric alone.* No priority claim; not refereed.

### 54. Surreal scales in quantum theory and gauge models

[physics/quantum-and-gauge-scale-reductions/article.pdf](#)

from four manuscripts, delivered 23 September 2026

What do exact infinitesimal scales add to finite quantum operations and gauge models, and what do they not? A rare branch of valuation  $2r$  has a conditional shadow given by a leading Gram matrix intrinsic to the output; after postselection with success probabilities  $p, q$  the trace distance is at most  $D/\max(p, q)$ , sharply, and in the classical case this halves the bound of **surreal/finite-surreal-probability**; two states can agree modulo every power of an infinitesimal yet have orthogonal conditional shadows. Separated virtual sectors are eliminated exactly, with independently separated

scales, support-certified block diagonalization, finite-depth hierarchies and Gibbs reduction, and soft-mode Schur defects are classified. For unit quaternions  $1 - \operatorname{Re} c = 2|u_1 \times u_2|^2$ , so regular gauge-invariant probes of a small loop start at second order; optimal  $m$ -query unambiguous identification of an infinitesimal holonomy succeeds with probability  $w_m = 1 - T_m(1 - w)$ ,  $T_m$  Chebyshev, and no finite test is perfect; a dark port heralds an exact  $\operatorname{SU}(2)$  unitary; and Hahn Kuranishi maps give charge rigidity and action gaps for gauge fields under imported Hodge–Green hypotheses. *Merged from four manuscripts that pairwise share theorems: each shared theorem is printed once crediting every proving source, and the claim-tier firewall of **physics/surreal-scalars-and-spacetime** is adopted (22 exact, 119 conditional, 119 assessments, 20 imports).* Not claimed: any departure from ordinary quantum mechanics or quantum field theory, a measurable infinitesimal, singularity resolution, or that **No** is physically necessary; every real specialization keeps the ordinary finite thresholds. Not refereed, no Lean formalization, priority not certified.

## 5 Foundations and computation

Nine reports that take the rest of the collection, or the surreal field as a set-theoretic object, as their object. The first three range over the whole subject; the others emphasize set theory, definability and computability, including a restricted exponential language for omnific arithmetic. Each of the first three is a merge of three manuscripts delivered together, the first two on 21 September 2026, and in those two the hazard was not overlap but a collision of near-identical names. The next two, on universes and birthday cutoffs, are merges of four and five manuscripts delivered on 22 September 2026, each proving one spine several times under different hypotheses. Two more, from batch 27, address definable surreals and omnific integers (two manuscripts) and omnific notations (three). The critical-point-defects report now integrates four manuscripts; the newest report treats exponential relations over omnific integers.

### 55. Foundations for surreal and surcomplex mathematics

[foundations-and-computation/foundations/article.pdf](#)

from [surreal\\_surcomplex\\_foundations.zip](#) + [surcomplex\\_foundations.zip](#) + [surreal\\_foundations.zip](#)

In what foundation does this mathematics legally live, and how would a proof assistant encode it? Compares ZF/ZFC with definable classes, GB/GBC/NBG, Kelley–Morse, Grothendieck universes and TG, Feferman reflection, constructive set theory, dependent type theory, univalent foundations and native Lean, and proves the size obstructions each must respect: no unrestricted cut operation can accept every subclass of its own carrier, the surreals are not Dedekind complete, and a universe cannot cut itself. The positive counterpart is a localization theorem — every *set* of surcomplex numbers already lies in one divisible set-sized workspace  $K_\Gamma = \mathbb{C}((t^\Gamma))$  — with a finite-algebra descent theorem explaining why a finite zero scheme gains no points in a larger workspace. Carries two series over  $\mathbb{C}((t^\mathbb{Q}))$  that differ only in the sign of an exponent and reach opposite conclusions:  $\sum t^{-n^2} X^n$  is summable at no nonzero argument, while  $\sum t^{+n^2} z^n$  is summable everywhere and is a *nonpolynomial* function coherent at every radius — which does not refute the all-scale rigidity theorem of **surcomplex/analysis**, stated for class functions on the whole **No**[ $i$ ], but shows it does not transfer to a fixed value group. *Merged; the danger was aggregation. All three manuscripts read the same Lean repository at the same revision on the same day, so their agreement is one act of source reading performed three times, not three confirmations. The findings are printed once and the verification boundary has its own early section.* The original merge compiled three illustrative Lean files and inspected their axiom dependencies. Those historical checks are distinct from the later development mapped in **FORMALIZATION.md**. The maintained article also makes the universe-relative localization hypotheses explicit, expands the positive-support recursion proof, and separates intrinsic convergence from full fine topology.

## 56. Surreal and surcomplex numbers in symbolic computer algebra

[foundations-and-computation/computer-algebra/article.pdf](#)

from `surreal_surcomplex_cas.zip` and two same-named archives

What can actually be computed exactly, and what only approximated or denoted? Separates finite exact denotation, effective coefficient access, decidable equality and order, and certified approximation into capability tiers, then works through an ordered rational-monomial core, real-algebraic closure, certified Hahn series, formal germs, analytic lifts, transseries, Newton profiles, Hensel lifting, finite quotient algebras, residues, and a proposed Wolfram Language architecture. *Merged; the danger was three prototypes with one name.* Each manuscript ships an executable Wolfram package, two of them near-anagrams of each other, two exporting six identical public symbol names with incompatible signatures, and all three calling themselves the rational monomial core. They are three different objects: coefficients  $\mathbb{Q}$  in one and  $\mathbb{Q}(i)$  in the others; exponent lattices  $\mathbb{Z}^d$  against  $\mathbb{Q}^r$ , so a ramified root is representable in one and not the next; and two of them order their monomial variables by opposite conventions, requiring the corresponding translation of monomial-order inequalities. The report never writes “the prototype” and tabulates the divergences instead. All three check suites were rerun for this repository and reproduce exactly — 34/34, 110/110 and 88/88 in Wolfram, 69/69 and 323/323 in Python — but one package cannot be loaded at all as shipped: an unclosed bracket stops it parsing, which is why its own record submits definitions rather than loading the file.

## 57. Computable surreal numbers, merged and reconciled

[foundations-and-computation/computable-surreals/article.pdf](#)

from `computable_surreal_numbers.zip` + two same-named archives

Which surreal numbers admit an algorithm, and in which representation? Three manuscripts arrived with two different numerical representations of the same objects — one using effective rational left-finite series with finite candidate covers, two using computable bounded-denominator Puiseux series — and the report is one integrated development rather than a concatenation: it preserves the distinct representations instead of picking one, supplies an explicit reconciliation ledger disposing of every divergence, keeps complementary proofs where the two routes prove the same statement by different means, and separates mathematical existence from the existence of a uniform algorithm throughout. *Merged; and this is the one report whose source manuscripts were never redistributed with it in the first place.* The ledger cites them by their original  $\text{\TeX}$  line numbers and `data/provenance-manifest.json` records their archive identities, file hashes and line counts, so every disposition is traceable against a copy obtained elsewhere. Four verification suites are rerun and pass — 4,967 exact comparisons, 18 named unit tests, 266 exact checks, and 4,366 exact cross-model and cover-contract checks in the updated reconciliation suite. The latter compare dense common-grid Puiseux convolution against sparse rational convolution, candidate covers with extra zeros and nonnested lists (including negative zero candidates in the geometric-series construction), the finite-jet root transformation, derivative and primitive identities, and finite instances of the two unbounded-denominator support examples. *The four units mean different things and are deliberately not added together.* A finite test proves no infinite theorem, no real closedness, no computability classification and no undecidability reduction; the synthetic halting fixtures in the original programs do not compute the halting set; the package implements neither arbitrary computable reals nor arbitrary infinite surreal arithmetic; and there is no Lean-checked formalization and no literature-wide priority claim.

## 58. Surreal fields across set-theoretic universes

[foundations-and-computation/surreal-fields-across-universes/article.pdf](#)

from four manuscripts, delivered 22 September 2026

How does the surreal field of an inner model look from a larger universe with the same ordinals? For transitive  $M \subseteq N$ , the old field  $\mathbf{No}^M$  is an elementary subfield of  $\mathbf{No}^N$ , yet its saturation, counted in  $N$ , can fail: for every uncountable cardinal  $\kappa$  of  $N$  it is externally  $\kappa$ -saturated exactly when  $N$  adds no ordinal-valued sequence of length below  $\kappa$ , and one small side of a cut already suffices;

at  $\omega$  the criterion is instead that no reals are added. A shortest new sequence, of regular length  $\delta$ , yields an explicitly decoded omitted cut of character  $(\delta, \delta)$ , and every externally set-presented gap is the trace of a unique fresh sign sequence all of whose proper prefixes are old. The first new birthday  $\beta$  satisfies  $\delta \leq \text{cf } \beta \leq \beta$ , and Cohen, closed and Prikry forcings are computed. The pure field  $\mathbf{No}^M[i]$  stays saturated over every set of parameters and, under Global Choice, is isomorphic to  $\mathbf{No}^N[i]$ ; naming conjugation restores the real spectrum, which gives real forms of one algebraically closed class field that are not conjugate to the standard one and have no set-sized cofinal subset. Outer sets of old numbers remain closed and uniformly discrete, and a set-complete class Boolean algebra in which reverse simplicity is dense answers a question from Berenbeim’s 2020 notes. *Merged from four manuscripts proving one spine; the interval-tree and sign-block-code constructions are kept as second routes, and results under plain ZFC are kept apart from those using Global Choice.* The real-form results continue, but do not settle, the real-form question of **surcomplex/surcomplex-field-automorphisms**. Three finite suites check sign codes and interval codes; a finite check proves no infinite theorem. Not refereed, no Lean verification, priority not certified.

## 59. Birthday cutoffs and hereditary sets

[foundations-and-computation/birthday-cutoffs-and-hereditary-sets/article.pdf](#)

from five manuscripts, delivered 22 September 2026

Which set universe does a bounded piece of surreal arithmetic remember? The ordered field is real closed and sees nothing, but once birthday is named, the surreals born below an epsilon number  $\lambda$  interpret the hereditary sets  $H_{\kappa(\lambda)}$ , with  $\kappa(\lambda)$  the least cardinal not below  $\lambda$ , and are bi-interpretable with  $(H_{\kappa(\lambda)}, \in, \lambda)$ ; already  $\mathbf{No}_{<\varepsilon_0}$  recovers every hereditarily countable set. Sign bits are read from order and birthday, a bounded ordinal pairing packs relations, and small certificates for Conway arithmetic avoid assuming Replacement in  $H_\kappa$ . Consequences: an exact classification of elementary inclusions between cutoffs, sentences for Power Set, Replacement and worldly cutoffs, the exact range of the coding without Choice, new-subset detectors and the first new birthday of an outer model, a Prikry extension that leaves a birthday-enriched field unchanged while singularizing its cutoff (from a measurable cardinal), and, for surcomplex cutoffs with conjugation and a coordinate birthday, the orientation obstruction and two lifts of every real elementary embedding. The full-class bi-interpretation with the set universe is the theorem announced by Chen–Hamkins–Yang, recorded in [foundations-and-computation/foundations](#); it is printed with a written proof as that announced theorem, not as a new result. *Merged from five manuscripts proving one spine at different cutoff classes; the most general is the base, and two arithmetic certificates and two set codings are kept as different proofs.* Five finite suites pass; a finite check proves no infinite theorem. Not refereed, no Lean verification, priority not certified.

## 60. Definable surreal numbers and omnific integers

[foundations-and-computation/definable-surreals-and-omnific-integers/article.pdf](#)

from two manuscripts, delivered 23 September 2026

Which surreal numbers and omnific integers are definable in the set-theoretic universe? For a set of parameters  $A$ , the  $A$ -definable surreals form a real closed field  $K_A$ , an elementary subfield of  $\mathbf{No}$  closed under the omega-map, exponential, logarithm and omnific floor, with its omnific integers as an integer part and  $K_A$  as their fraction field. The ordinal-definable surreals are exactly  $\mathbf{No}^{\text{HOD}}$ , and  $V = \text{HOD}$  holds iff every omnific integer in  $(0, \omega)$  is ordinal definable, iff every monomial  $\omega^a$  with  $0 < a < 1$  is: parameter-free reversible codes place every surreal in a bounded omnific interval. Otherwise an Ord-indexed algebraically independent family lies over the definable field. In a transitive model the first non-definable ordinal is an epsilon number and bounds the maximal initial core, an elementary exponential subfield with an omnific integer part; the first birthday of a non-OD surreal is the first ordinal with a non-HOD subset; forcing over  $L$  gives the examples. *Merged from two manuscripts; the base has the weaker hypotheses.* Not claimed: codes that are ring maps or definable in the ordered field, a truth predicate for  $V$ , definability as computability. Proof review pending; not refereed, no Lean formalization, priority not certified.

## 61. Notations for omnific integers

[foundations-and-computation/omnific-notations/article.pdf](#)

from three manuscripts, delivered 23 September 2026

How can omnific integers be named exactly, and at what cost? Three countable notation systems: holonomic fraction towers, whose fields have decidable equality and order, effective truncation, a computable floor and supports of order type at most  $\omega^r$ , with an  $\varepsilon_0$ -labelled union; hereditary Hahn forms with canonical trees and decidable omnific membership and floor, extended by shielding to effective coefficient rings; and rational  $\omega$ -terms with decidable validity, equality, order, membership in **Oz** and coefficients, whose ordinals are exactly those below  $\varepsilon_0$ . For raw names with  $d$  blocks, equality is  $\Pi_1^0$ -complete, there is no translator into any system with decidable equality, positivity is complete for the  $d$ -c.e. sets, and support validity is  $\Pi_1^1$ -complete. *Merged from three manuscripts; the base's hypotheses, any effective ordered coefficient field and  $d$  blocks, are the weakest.* Not claimed: a notation for all of **Oz**, real closedness, a uniform complexity bound, an answer to a question of [foundations-and-computation/computable-surreals](#). Proof review pending; not refereed, no Lean formalization, priority not certified.

## 62. Critical-point defects in surreal arithmetic

[foundations-and-computation/large-cardinal-embeddings-and-normal-forms/article.pdf](#)

from four manuscripts, through fc57b61, 23 September 2026

How does an elementary embedding  $j : V \rightarrow M$  with critical point  $\kappa$  act on surreal numbers? Applying  $j$  to sign sequences and applying it termwise to the exponents of a Conway normal form are different maps. The report claims that they agree exactly on normal forms with fewer than  $\kappa$  terms, that their difference is an explicit subseries, and that the defect of a Boolean mask records membership in the normal measure derived from  $j$ , with omnific and surcomplex versions. The companions add criteria for descent to the target model, support covers, omnific algebraic independence and measurable first failures. A measurable cardinal supplies an instance of the hypotheses; its existence is not asserted. Independent proof review and source reconciliation remain pending. Not refereed, no Lean formalization, priority not certified.

## 63. Exponential relations over omnific integers

[foundations-and-computation/exponential-relations-over-omnific-integers/article.pdf](#)

from one manuscript, placed in aa9c891, 23 September 2026

Which finite exponential relations see only ordinary integers and purely infinite coordinates? On the domain  $\overline{\mathbb{Q}} + J \subseteq \mathbf{No}[i]$ , where  $J$  is the purely infinite part, the report combines ordinary complex and Gonshor exponentials to obtain exact toric relations. With algebraic coefficients, algebraic shifts and real-algebraic slopes, it proposes a decidable additive language of exponential-equality predicates. Its elementary cores are  $\mathbb{Z} + W$  for nonzero ordered vector subspaces  $W \subseteq J$  over the real algebraic numbers. A separate fixed irrational slope has the Turing degree of its rational cut; admitting  $\exp(xy) = \exp(z)$  gives undecidability. This does not define an exponential on all surcomplex numbers or decide the unrestricted surreal exponential field. Independent proof review and Lean formalization remain pending.